AutoSAM: Auto-Prompting Mamba-Based Vision Foundation Model for Multimodal Remote Sensing Semantic Segmentation

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***Abstract*—Vision foundation models, such as the segment anything model (SAM), have advanced remote sensing (RS) tasks. However, extending SAM to multimodal RS semantic segmentation faces two key challenges: 1) SAM is tailored for unimodal inputs and lacks RS-specific knowledge, hindering effective spatial modeling and cross-modal feature integration; and 2) SAM depends on externally provided prompts (e.g., points or boxes), limiting its scalability and practicality in multimodal scenarios. To address these issues, we present AutoSAM, an end-to-end auto-prompting Mamba-based vision foundation model framework tailored for multimodal RS semantic segmentation. Specifically, we introduce a CrossMamba block (CMB) in the feature extraction stage to replace the conventional multihead self-attention mechanism, where the core reverse interactive scanning adaptor-SS2D module (RISASM) promotes semantic interaction and alleviates modality discrepancies. In addition, a multimodal scale-aware fusion module (MSAFM) is incorporated to enhance scale-aware fusion and suppress irrelevant features through cascaded residual interactions. Furthermore, we propose a plug-and-play multimodal mixture-of-class-expert auto-prompting module (MMoEAPM), which enables the generation of pseudo-mask prompts for the original prompt encoder without additional training overhead, thereby supporting efficient auto-prompting. Extensive experiments and ablation studies on four benchmark multimodal RS datasets demonstrate that AutoSAM consistently achieves state-of-the-art performance across diverse modality combinations. The code is available at https://github.com/NWPUFranklee/AutoSAM.git**

***Index Terms*—Auto-prompting strategy, multimodal remote sensing (RS), semantic segmentation, vision foundation model.**

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I. Introduction

MULTIMODAL remote sensing semantic segmentation (MRSSS) aims to accurately assign semantic categories to each pixel by leveraging complementary information from multisensor images of the same scene [1], [2], [3]. This task plays a crucial role in land cover monitoring [4], Earth observation [5], and defense applications [6]. With the rapid advancement of artificial intelligence, unimodal remote sensing (RS) segmentation models have achieved remarkable progress, yet their performance has reached a plateau [7], [8], [9]. Building on state-of-the-art visual segmentation architectures such as UNet [10] (CNNs-based), Segformer [11] (transformer-based), and SegMAN [12] (Mamba-based), recent research has explored modality-specific strategies for diverse RS data. Nevertheless, extensive experiments have shown that integrating multimodal features for complementary advantage consistently leads to superior ground perception compared to unimodal approaches [13], [14], [15].

To advance multimodal RS, substantial efforts have been devoted to developing domain-specific datasets [16], [17] and advanced models [18], [19], [20], [21]. Nevertheless, existing approaches often suffer from limited cross-modality generalization, poor scalability, restricted user interaction, and suboptimal segmentation accuracy. A notable recent development in vision foundation models is the segment anything model (SAM) [17], which adopts a prompt-driven paradigm for semantic segmentation. Pretrained on the large-scale SA-1B natural image dataset, SAM is able to capture rich visual contextual information and effectively integrate user-provided prompts, thereby enabling strong zero-shot generalization. This paradigm shift has facilitated the application of foundation models across diverse fields, including medical image segmentation. Its potential in RS has also attracted increasing interest. For example, SAM-RSIS [22] employs a progressive fine-tuning strategy with box prompts to transfer segmentation capabilities to RS instance segmentation, while RS-SAM [23] introduces a multiscale neck and progressive refinement modules to address the challenges of segmenting small objects and complex backgrounds. Given the demonstrated effectiveness of SAM on unimodal RS segmentation, recent studies have increasingly investigated its extension to multimodal RS segmentation. These efforts aim to leverage

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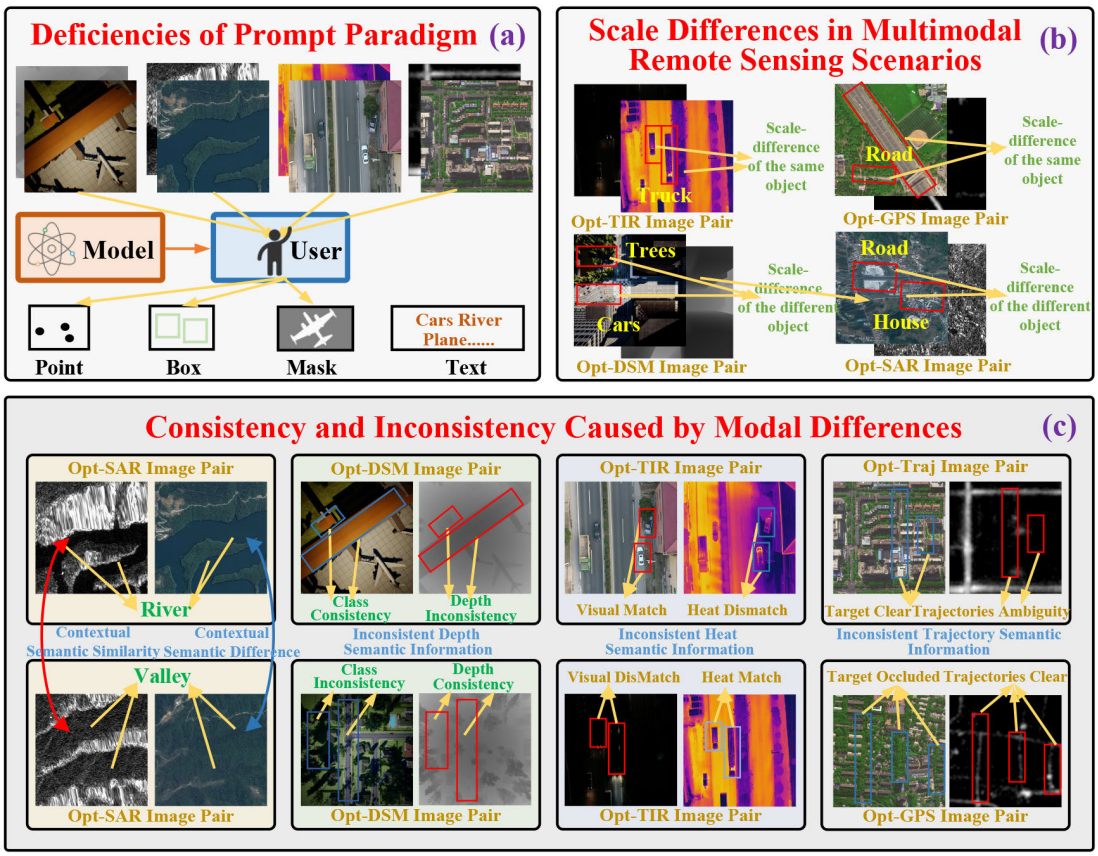


Fig. 1. Illustration of key challenges in multimodal RS semantic segmentation: (a) limitations and inefficiencies of prompt-based segmentation paradigms that rely on manual input, (b) significant scale discrepancies both within and across object categories in different modalities, and (c) consistency and inconsistency issues caused by modality differences, including semantic, spatial, and contextual mismatches.

SAM's powerful prompt-driven capabilities for integrating heterogeneous data sources.

However, several critical challenges persist in applying SAM-based methods to MRSSS tasks as follows.

1) As illustrated in Fig. 1(a), the prompt encoder is a core component that enables SAM to achieve high-precision segmentation. Nevertheless, SAM still relies on externally provided manual prompts. Some medical imaging studies even utilize label-derived point and box prompts for training. This dependence on external prompts is both inefficient and resource-intensive, and is further exacerbated in high-resolution and complex RS scenarios, thereby significantly limiting the scalability and practical deployment of SAM in RS semantic segmentation applications.
2) As shown in Fig. 1(b), SAM lacks an inherent ability to capture scale in multimodal data. Although recent works have attempted to incorporate multiscale modules into SAM, these solutions often face compatibility issues with SAM's architecture. Scale variation is particularly critical in multimodal RS, as it typically covers vast spatial extents and contains objects with significant scale disparities. This makes direct application of SAM challenging, since scale differences can occur both between and within segmentation targets.
3) As depicted in Fig. 1(c), notable modality gaps exist among different types of RS data, such as optical, SAR, and TIR images. These differences are reflected in feature distributions, spectral responses, and semantic representations, posing challenges for feature alignment and label consistency. For example, the same object may exhibit distinct characteristics in optical and SAR images, while thermal and visual information in TIR and optical data can be inconsistent under varying conditions. Such discrepancies hinder the direct application of SAM-based models in multimodal tasks.
4) In addition, the transformer-based multihead self-attention mechanism of SAM is adept at modeling contextual features within a single modality, but struggles to effectively integrate semantic information across modalities. Although the self-attention reduces sequence length via window partitioning, it retains quadratic complexity and may lead to the loss of global information. This underscores the need for more efficient and effective strategies for global and cross-modal feature modeling in multimodal RS semantic segmentation.

To address the aforementioned challenges, such as large modality gaps, difficulty of feature alignment and fusion, and limited generalization and interpretability, in multimodal RS semantic segmentation, we propose an end-to-end SAM-based

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auto-prompting vision foundation model, termed AutoSAM. Specifically, AutoSAM is built upon the linear-complexity VMamba [24] backbone and introduces the CrossMamba block (CMB) to replace the conventional multihead self-attention mechanism [25], thereby effectively bridging large modality gaps and improving cross-modal semantic understanding. Within the CMB, the core reverse interactive scanning adaptor-SS2D module (RISASM) enables multidirectional, reverse information exchange between modalities, explicitly mitigating semantic and contextual discrepancies. To address the difficulty of feature alignment and fusion across heterogeneous modalities and varying object scales, AutoSAM incorporates the multimodal scale-aware fusion module (MSAFM) and RISASM in a global–local feature enhancement structure. The MSAFM adaptively fuses features from different modalities at multiple scales, preserving salient information and suppressing irrelevant features for robust and precise alignment. Furthermore, to enhance generalization and interpretability while overcoming the manual dependency of prompting in SAM, we design a plug-and-play self-prompting strategy, the multimodal mixture-of-class-expert auto-prompting module (MMoEAPM), which combines class correlation dictionary learning with a mixture-of-experts (MoEs) [26] framework to generate high-quality pseudo-labels, supporting both self-prompting and manual prompting, thus enabling flexible adaptation to various multimodal scenarios. The main contributions of this work are summarized as follows.

1) We propose, for the first time, a unified vision foundational model for multimodal RS semantic segmentation, AutoSAM, which supports both automatic and manual prompting. Extensive experiments on four modality combinations (Optical-SAR, Optical-DSM, Optical-TIR, and Optical-Traj) demonstrate its superior segmentation performance.
2) We design a multimodal category-relevance-aware architecture, MMoEAPM, based on MoE-Dict, which leverages deep decoding of multimodal branch features to fully utilize multimodal information and endow SAM with efficient, plug-and-play self-prompting capabilities.
3) AutoSAM incorporates the RISASM within the proposed CMB, marking the first replacement of the ViT-based encoder with a VMamba-based structure in SAM for multimodal RS semantic segmentation. This design substantially enhances cross-modal semantic understanding and alleviates modality gaps.
4) To address scale variation in multimodal RS semantic segmentation, we introduce the MSAFM within the CMB, establishing a global–local feature extraction architecture that enables SAM to model multiscale correlations and adaptively reweight features from multiple RS modalities.

The article is structured as follows. Section II reviews the related work. Section III presents the proposed AutoSAM framework and its associated components. Section IV shows the experimental analysis, and Section V provides the discussion. Section VI concludes the work.

II. Related Works

A. Auto-Prompting Strategy for SAM

Recent advances have focused on reducing SAM's reliance on manual prompts by developing various auto-prompting strategies across diverse domains. AdapSAM [27] adapts SAM for medical image segmentation by introducing an auxiliary prompt encoder that directly uses the input image as a prompt, without fine-tuning the core SAM model. SAM-SP introduces the self-prompt fine-tuning approach [28], enabling SAM to operate in specific domains without manual prompts by generating self-prompts from multiscale features and optimizing segmentation via dynamic mask selection and label refinement. For the RS field, TPP-SAM [29] addresses zero-shot road extraction by leveraging GPS trajectory points as prompts, eliminating the need for complex manual annotations. Self-Prompt-SAM [30] utilizes a multiscale prompt generator combined with the image encoder to automatically generate auxiliary masks, extracting bounding boxes and center points as prompts, thus adapting 2-D SAM for image segmentation. ST-SAM [31] leverages SAM-based pseudo-labeling and self-training to boost semi-supervised camouflaged object detection efficiency. In weakly supervised settings, APSAM [32] automatically generates mixed prompts (bounding boxes and points) through an object localization network and employs an adaptive prompt generation algorithm to fully exploit class activation map patterns for landslide extraction. AoP-SAM [33] leverages a learnable prompt predictor to generate optimal point prompts from image embeddings and introduces an adaptive sampling and filtering mechanism to reduce redundant mask generation. PTSAM [34] employs parameter-efficient fine-tuning, allowing SAM to adapt to specific domains with only 2048 additional parameters and as few as 16 training images. AutoProSAM [35] facilitates seamless adaptation from 2-D natural images to 3-D multiorgan segmentation via an automatic prompt generator. In the geographic RS domain, GeoSAM [36] enhances SAM's segmentation of transportation infrastructure by integrating multimodal prompts and employing parameter-efficient fine-tuning. These methods collectively highlight the trend toward automated, domain-adaptive prompting for SAM, significantly reducing the reliance on manual annotations and enabling more efficient and versatile segmentation.

B. Parameter-Efficient Fine-Tuning Methods

Recent research has explored parameter-efficient strategies to adapt foundation models such as SAM for multimodal RS semantic segmentation. Yan et al. [37] propose RingMo-SAM, a foundational model capable of segmenting and recognizing object categories in both optical and SAR images. Pan et al. [38] introduce VSGNet, which leverages visual structural guidance from pretrained models like SAM and fuses complementary information from different sensors via a cross-modal collaborative network that combines CNN and ViT architectures, effectively addressing imprecise multisource data registration. Zhang et al. [39] present FlexiSAM, a flexible segmentation model tailored for high-resolution multimodal land cover classification, integrating RGB and near-infrared

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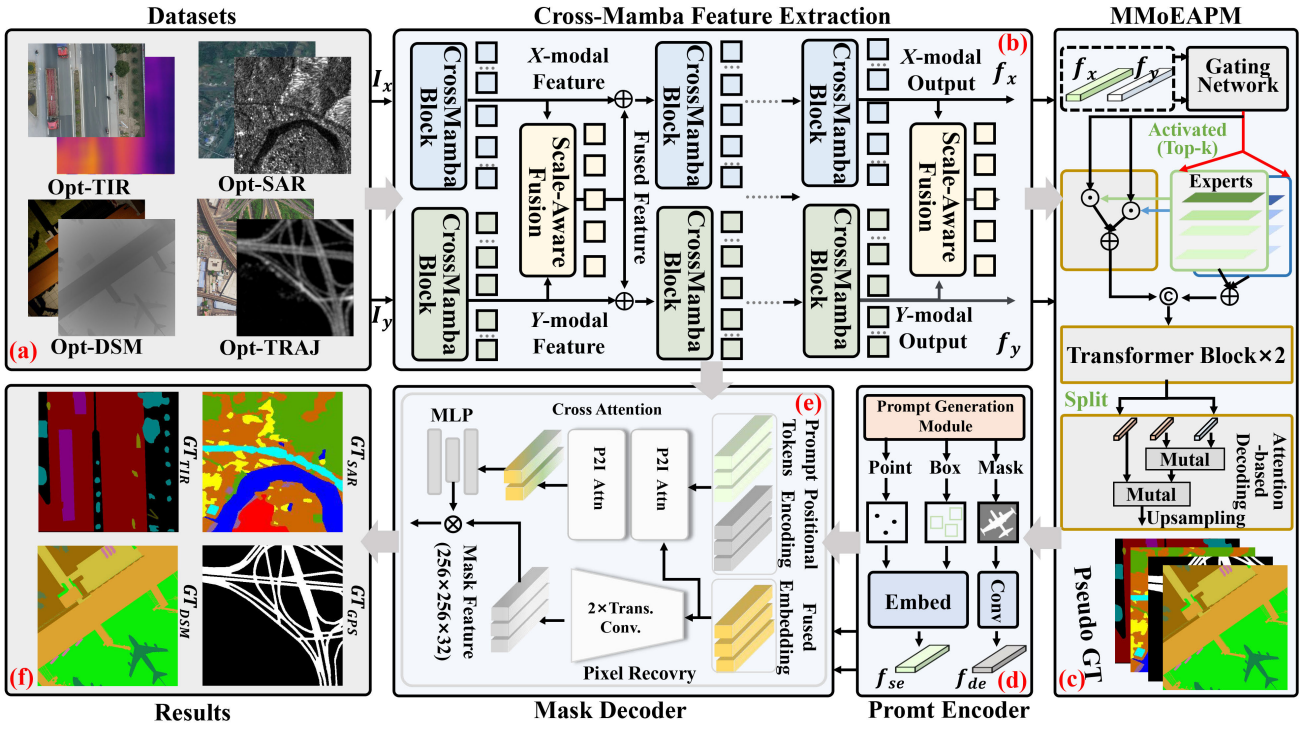


Fig. 2. Overall architecture of the proposed AutoSAM framework. (a) Example inputs from four representative multimodal RS datasets, including optical-thermal (Opt-TIR), optical-SAR (Opt-SAR), optical-DSM (Opt-DSM), and optical-trajectory (Opt-TRAJ) pairs. (b) CrossMamba feature extraction block (CMB) extracts global features from each modality and performs scale-aware fusion. (c) MMoEAPM adaptively generates pseudo-segmentation masks via an MoEs strategy and a class dictionary. (d) Prompt encoder extracts sparse (point, box) and dense (mask) embeddings from the pseudo masks for downstream processing. (e) Mask decoder aggregates the scale-fused features and prompt embeddings to produce the final segmentation results. (f) Visualizations of ground-truth and predicted segmentation maps on representative datasets. The entire pipeline enables end-to-end, adaptive, and efficient multimodal segmentation with minimal manual intervention.

data to better delineate complex land boundaries. VTPSeg [40] is an open-vocabulary semantic segmentation framework that integrates grounding DINO, CLIP, and SAM, mitigating issues such as redundant mask generation in SAM by utilizing explicit prompts. SkySense-O [41] integrates vision-centric contrastive learning and visual-language pretraining for open-world RS interpretation. Bi et al. [42] propose ViRefSAM, which adopts a few-shot learning approach by using a small set of annotated reference images to guide SAM, thereby reducing reliance on manual prompting and improving domain adaptability. Wu et al. [43] proposed a deep convolutional neural network framework for multimodal RS data classification, which utilizes advanced cross-channel reconstruction to enhance feature fusion across modalities. Furthermore, Zhang et al. [44] exploit SAM's feature extraction capabilities by integrating SAR and DEM data for flood area segmentation. Li et al. [45] introduced SeaMo, a season-aware multimodal foundation model that leverages masked image modeling and multiseasonal fusion to improve generalization and adaptability in RS tasks. Building on SAM2, Zhu et al. [46] introduce EarthMind, a unified framework for multigranularity and multisensor Earth observation data processing. Gao et al. [47] develop SFA-Net, which utilizes SAM-extracted edge structures as priors to enhance local feature distinctiveness for multimodal image matching. Tao et al. [48] improve SAM for farmland parcel extraction by introducing automatic prompt generation modules, enabling precise segmentation from multisource data. Ma et al. [20] propose MFNet, which fine-tunes SAM with MMAdapter to adapt to semantic segmentation tasks involving RGB and DSM images. Collectively, these works demonstrate the effectiveness of parameter-efficient fine-tuning and cross-modal adaptation in extending the capabilities of SAM for complex multimodal RS scenarios.

III. Methodology

As illustrated in Fig. 2, we present a detailed overview of AutoSAM and its core components. AutoSAM is a novel end-to-end foundational model designed for multimodal semantic segmentation. The framework primarily consists of three key modules: a global interactive feature extraction module based on the CMB, which facilitates semantic understanding and information exchange between multimodal features; a plug-and-play MMoEAPM, which leverages an MoEs architecture and class dictionary learning to automatically generate high-quality pseudo-segmentation masks for self-prompting; and a global–local feature extraction and fusion framework realized through the MSAFM, which promotes effective multiscale fusion and aggregation of multimodal features. These components work synergistically to enable AutoSAM to adaptively model complex relationships across modalities, generate reliable prompts without manual intervention, and achieve accurate semantic segmentation in diverse RS scenarios.

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A. Overall Structure of AutoSAM

The original SAM framework employs a pretrained ViT [49] as its image encoder backbone. To better preserve multimodal information integrity and improve computational efficiency, we replace the transformer block with the CMB, denoted as $\mathcal{F}^{(i)}_{\text{CMB}}(\cdot\,|\,\boldsymbol{\theta})$, as shown in Fig. 2(b). This module retains the original hyperparameter settings while enabling more efficient global feature extraction. where i denotes the ith CMB layer and $\boldsymbol{\theta}$ represents its learnable parameters. Given a multimodal input image $\mathbf{I}_k \in \mathbb{R}^{3\times H\times W}$, where $k \in \{\text{x}, \text{y}\}$ indexes the modality, we first feed it into the first CMB layer to obtain the initial global feature $\mathbf{F}^{(1)}_k$. At each subsequent layer, the extracted global features are aggregated with the scale-fused features $\mathbf{F}^{(i)}_\Delta$ for global–local fusion, yielding optimized modality-specific features $\mathbf{F}^{(i+1)}_k$. The scale-fused feature for the first layer is initialized as zero, i.e., $\mathbf{F}^{(1)}_\Delta = \mathbf{0}$, which is formulated as

$$\mathbf{F}^{(i+1)}_k = \mathcal{F}^{(i)}_{\text{CMB}}\left(\mathbf{F}^{(i)}_k + \mathbf{F}^{(i)}_\Delta\,|\,\boldsymbol{\theta}\right). \tag{1}$$

To further enhance multimodal fusion and capture multiscale details, we introduce an MSAFM, denoted as $\mathcal{F}^{(i)}_{\text{MSAF}}(\cdot,\cdot\,|\,\boldsymbol{\theta})$, after each CMB layer. MSAFM receives global features from different modalities at the same layer and produces the scale-fused features $\mathbf{F}^{(i)}_\Delta$ as follows:

$$\mathbf{F}^{(i)}_\Delta = \mathcal{F}^{(i)}_{\text{MSAF}}\left(\mathbf{F}^{(i)}_k, \mathbf{F}^{(i)}_{\Delta\backslash k}\,|\,\boldsymbol{\theta}\right) \tag{2}$$

where $\mathbf{F}^{(i)}_{\Delta\backslash k}$ denotes the aggregated features from modalities other than k. As illustrated in Fig. 2(c), the MMoEAPM, formulated as $\mathcal{F}_{\text{MMoEAPM}}(\cdot,\cdot\,|\,\boldsymbol{\theta})$, serves as the core for prompt strategy selection. It receives the final optimized features $\mathbf{F}_k$ and $\mathbf{F}_{\Delta\backslash k}$, and generates a pseudo-segmentation mask $\hat{\mathbf{Y}}$ as follows:

$$\hat{\mathbf{Y}} = \mathcal{F}_{\text{MMoEAPM}}\left(\mathbf{F}_k, \mathbf{F}_{\Delta\backslash k}\,|\,\boldsymbol{\theta}\right). \tag{3}$$

This pseudo mask is used for subsequent self-prompting and is supervised by ground-truth labels through backpropagation. For effective prompt representation, we retain the original SAM prompt encoder, denoted as $\mathcal{F}_{\text{PE}}(\cdot\,|\,\boldsymbol{\theta})$, as shown in Fig. 2(d). This module processes $\hat{\mathbf{Y}}$ to extract sparse (point, box) embeddings $\mathbf{f}_{\text{se}}$ and dense (mask) embeddings $\mathbf{f}_{\text{de}}$

$$(\mathbf{f}_{\text{se}},\, \mathbf{f}_{\text{de}}) = \mathcal{F}_{\text{PE}}\left(\hat{\mathbf{Y}}\,|\,\boldsymbol{\theta}\right). \tag{4}$$

Finally, the last scale-fused feature $\mathbf{F}^{(i)}_\Delta$ together with the prompt vectors $\mathbf{f}_{\text{se}}$ and $\mathbf{f}_{\text{de}}$ are fed into the original mask decoder $\mathcal{F}_{\text{MD}}(\cdot,\cdot,\cdot\,|\,\boldsymbol{\theta})$ to generate the final segmentation result $\mathbf{Y}$, as illustrated in Fig. 2(e)

$$\mathbf{Y} = \mathcal{F}_{\text{MD}}\left(\mathbf{F}^{(i)}_\Delta, \mathbf{f}_{\text{se}}, \mathbf{f}_{\text{de}}\,|\,\boldsymbol{\theta}\right). \tag{5}$$

AutoSAM jointly leverages efficient global–local feature extraction, adaptive prompt generation, and multiscale feature fusion, enabling robust and accurate semantic segmentation. The specific calculation process is shown in Algorithm 1.

Algorithm 1 Inference Pipeline of AutoSAM

Input: Multimodal images $\mathbf{I}_x, \mathbf{I}_y$
Output: Segmentation mask $\mathbf{Y}$

$\mathbf{F}^{(0)}_x \leftarrow \mathbf{I}_x$; $\mathbf{F}^{(0)}_y \leftarrow \mathbf{I}_y$; // Initialize modality-specific features
$\mathbf{F}^{(1)}_\Delta \leftarrow \mathbf{0}$; // Initialize scale-fused feature
for $i = 1$ **to** N **do**
  $\mathbf{F}^{(i)}_x \leftarrow \mathcal{F}^{(i)}_{\text{CMB}}\left(\mathbf{F}^{(i-1)}_x + \mathbf{F}^{(i)}_\Delta\,|\,\boldsymbol{\theta}\right)$; // X-modality feature extraction
  $\mathbf{F}^{(i)}_y \leftarrow \mathcal{F}^{(i)}_{\text{CMB}}\left(\mathbf{F}^{(i-1)}_y + \mathbf{F}^{(i)}_\Delta\,|\,\boldsymbol{\theta}\right)$; // Y-modality feature extraction
  $\mathbf{F}^{(i+1)}_\Delta \leftarrow \mathcal{F}^{(i)}_{\text{MSAF}}\left(\mathbf{F}^{(i)}_x, \mathbf{F}^{(i)}_y\,|\,\boldsymbol{\theta}\right)$; // Multi-scale fusion
$\mathbf{F}_x \leftarrow \mathbf{F}^{(N)}_x$; $\mathbf{F}_y \leftarrow \mathbf{F}^{(N)}_y$; // Final modality features
$\hat{\mathbf{Y}} \leftarrow \mathcal{F}_{\text{MMoEAPM}}(\mathbf{F}_x, \mathbf{F}_y\,|\,\boldsymbol{\theta})$; // Class-expert prompt generation
$(\mathbf{f}_{\text{se}}, \mathbf{f}_{\text{de}}) \leftarrow \mathcal{F}_{\text{PE}}(\hat{\mathbf{Y}}\,|\,\boldsymbol{\theta})$; // Sparse/dense prompt embedding
$\mathbf{Y} \leftarrow \mathcal{F}_{\text{MD}}(\mathbf{F}^{(N+1)}_\Delta, \mathbf{f}_{\text{se}}, \mathbf{f}_{\text{de}}\,|\,\boldsymbol{\theta})$; // Segmentation prediction
return $\mathbf{Y}$

B. CrossMamba Block

As illustrated in Fig. 3(a), the CMB is the first core module in the feature extraction stage of AutoSAM. The CMB not only inherits the responsibility of global feature modeling from the transformer block, but also plays a critical role in mitigating modal discrepancies. Specifically, CMB is inspired by VMamba [24], which achieves long-range dependency modeling comparable to transformers through a sequence scanning mechanism, while its linear computational complexity significantly enhances real-time performance. Given the kth modality input feature $\mathbf{X}_k$, the CMB first applies layer normalization, denoted as $\mathcal{LN}(\cdot)$, followed by contextual modeling via the cross-SS2D operation $\mathcal{CSS}\in\mathcal{D}(\cdot)$. The resulting intermediate feature $\mathbf{Z}_k$ is obtained by a residual connection operation as

$$\mathbf{Z}_k = \mathcal{CSS}\in\mathcal{D}\left(\mathcal{LN}\left(\mathbf{X}_k\right)\right) + \mathbf{X}_k. \tag{6}$$

The normalized intermediate feature $\mathcal{LN}(\mathbf{Z}_k)$ is then forwarded to a feedforward network $\mathcal{FFN}(\cdot)$ with another residual connection to yield the CMB's output

$$\mathbf{X}^{\text{out}}_k = \mathcal{FFN}\left(\mathcal{LN}\left(\mathbf{Z}_k\right)\right) + \mathbf{Z}_k \tag{7}$$

where $\mathbf{X}^{\text{out}}_k$ represents the updated feature for modality k. The core computation of the cross-SS2D block is defined as

$$\mathbf{X}^{\text{out}}_k = \mathcal{LN}\left(\mathcal{R}\left(\text{LDS}\left(\mathbf{X}_k\right), \text{LDS}\left(\mathbf{X}_k\right)\right)\right) \otimes \mathcal{S}\left(\mathbf{X}_{\Delta\backslash k}\right) \tag{8}$$

where $\mathcal{LDS}(\cdot)$ denotes a linear layer followed by depthwise convolution and SiLU activation; $\mathcal{LN}(\cdot)$ is a linear layer with layer normalization; $\mathcal{S}(\cdot)$ denotes the SiLU activation function; $\mathbf{X}_{\Delta\backslash k}$ indicates features from modalities other than k; $\mathcal{R}(\cdot)$ denote the reverse interactive scanning adaptor SS2D module (RISASM). Compared with conventional transformer-based or convolutional backbones, the proposed CMB leverages the state-space modeling ability of VMamba and integrates explicit cross-modal feature exchange via RISASM, enabling

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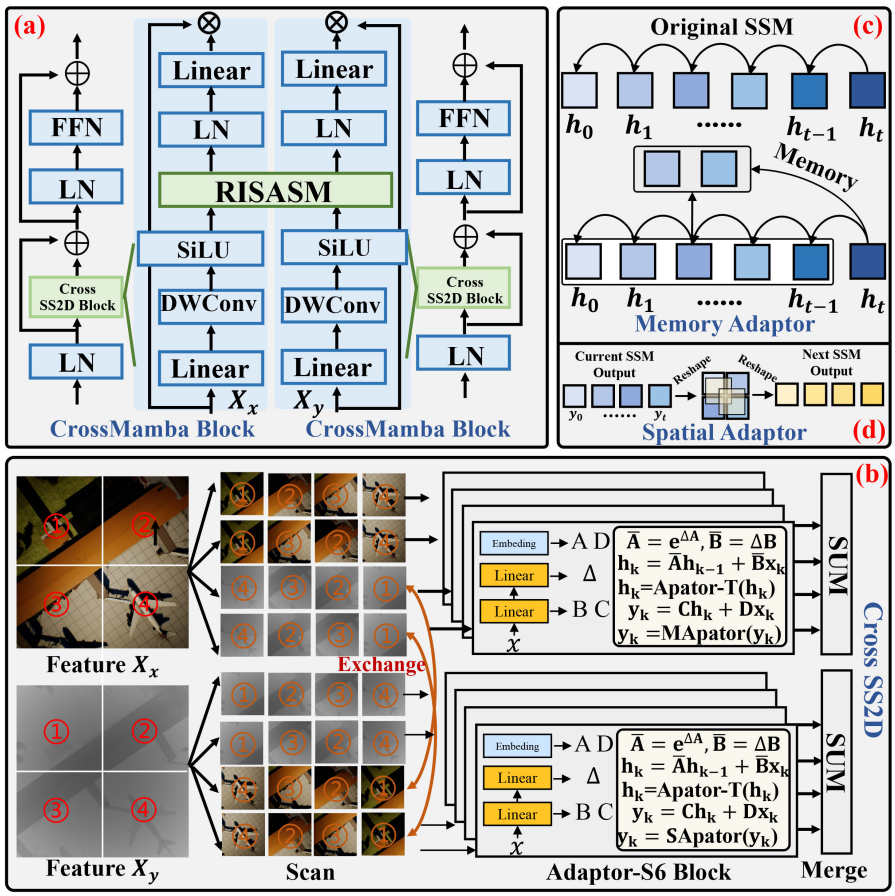


Fig. 3. Architecture of the proposed CMB and its key components. (a) Overview of the CMB, which integrates the cross-SS2D operation and the RISASM to achieve global feature modeling and cross-modal interaction. (b) Illustration of the cross-SS2D block, including four-directional feature scanning, exchange between modalities in the reverse directions, and the use of adaptor-S6 blocks for context aggregation. (c) Original S6 block based on the SSM, showing the sequential memory update mechanism for capturing long-range dependencies. (d) Proposed adaptor-S6 block, which introduces a memory adaptor to recover forgotten hidden states and a spatial adaptor to restore local spatial information, thus enhancing both long-range and spatial modeling capabilities. The overall design enables efficient and robust multimodal feature extraction by combining sequence modeling, adaptive memory retrieval, and spatial context recovery.

more efficient global context modeling and cross-modal semantic interaction for multimodal RS.

C. Reverse Interactive Scanning Adaptor SS2D Module

As shown in Fig. 3(b), the RISASM, denoted as $\mathcal{R}$, serves as the core computational unit within the CMB and is fundamental for multimodal feature contextual modeling. RISASM extends the SS2D module by inheriting its four-directional scanning strategy, which compensates for the loss of 2-D spatial information in the Mamba framework and supports richer global feature modeling. Given the kth modality input feature $\mathbf{X}_k$, RISASM first applies the directional scan operation $\mathcal{SCAN}(\cdot)$ to obtain four scanning features $\mathbf{S}_k^{(i)}$, where $i \in \{1, 2, 3, 4\}$ are the scanning direction

$$\mathbf{S}_k^{(i)} = \mathcal{SCAN}\left(\mathbf{X}_k\right), \quad i \in \{1, 2, 3, 4\}. \tag{9}$$

To enable effective cross-modal information integration, RISASM introduces a reverse interactive scanning mechanism. Specifically, for the last two scanning directions ($i = 3, 4$), RISASM performs a cross-modal feature exchange, i.e., the scanning features of the two modalities are swapped as

$$\mathbf{S}_k^{(j)}, \mathbf{S}_{\Delta\backslash k}^{(j)} = \mathbf{S}_{\Delta\backslash k}^{(j)}, \mathbf{S}_k^{(j)}, \quad j \in \{3, 4\}. \tag{10}$$

This reverse exchange allows each modality to access and incorporate both local and global contextual cues from the other modality, thereby enriching the semantic representation and improving the model's ability to capture cross-modal dependencies. The design facilitates mutual information flow and enhances reverse semantic interpretability in multimodal feature extraction. Each scanning feature $\mathbf{S}_k^{(i)}$ is subsequently processed by an adaptor-S6 block, denoted as $\mathcal{AS}(\cdot)$, resulting in direction-aware global features

$$\mathbf{A}_k^{(i)} = \mathcal{AS}\left(\mathbf{S}_k^{(i)}\right). \tag{11}$$

The outputs from all directions are finally aggregated to obtain the output of RISASM for the kth modality

$$\mathbf{X}_k^{\text{out}} = \sum_{i=1}^{4} \mathbf{A}_k^{(i)}. \tag{12}$$

The reverse interactive scanning mechanism introduces only a minor increase in computational cost, as the cross-modal feature exchange is implemented via simple tensor swapping and aggregation, which are highly parallelizable and efficient. Thus, the proposed mechanism strikes an effective balance between cross-modal expressiveness and efficiency.

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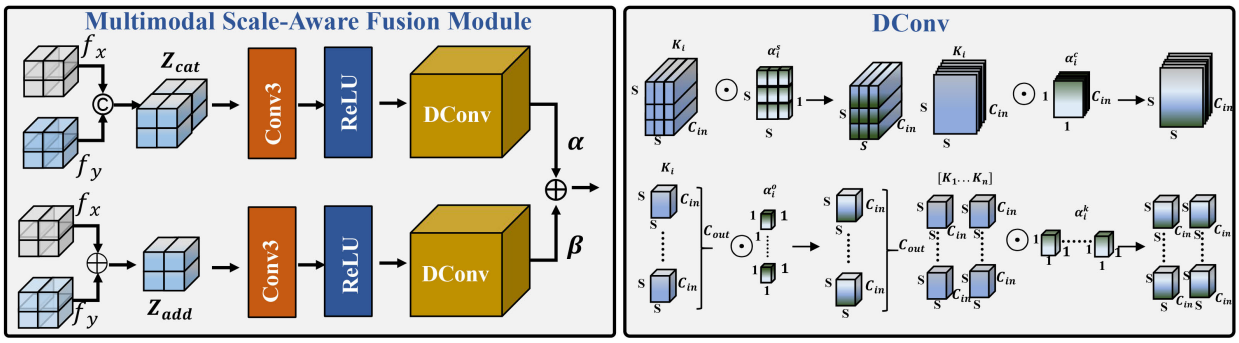


Fig. 4. Illustration of the MSAFM. The module fuses features from different modalities using both concatenation and addition operations, followed by DConv layers. The outputs of both branches are adaptively combined via learnable weights. The right panel details the DConv process, where convolution kernels are dynamically modulated along multiple dimensions to achieve multiscale feature fusion.

D. Adaptor-S6 Block

The adaptor-S6 block is designed to address the limitations of the standard S6 block, particularly the issues of long-range forgetting and spatial information loss in sequential modeling. The original S6 block converts the continuous-time state-space model (SSM) parameters $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ into their discrete equivalents $(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \bar{\mathbf{C}}, \mathbf{D})$ via zero-order hold (ZOH), enabling sequence modeling as follows:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{u}_t, \qquad \mathbf{y}_t = \bar{\mathbf{C}}_t \mathbf{h}_t + \mathbf{D}\mathbf{u}_t. \tag{13}$$

However, this recursive paradigm is prone to forgetting long-range dependencies, which is detrimental to modeling complex spatial relationships in RS imagery. To mitigate this, as shown in Fig. 3(c) and (d), we insert two key adaptors.

1) Memory Adaptor: After each recursive step, a memory adaptor $\mathcal{M}(\cdot)$ is introduced to recover information from forgotten hidden states as follows:

$$\mathbf{h}_t = \mathcal{M}\left(\bar{\mathbf{A}}_t \mathbf{h}_{t-1} + \bar{\mathbf{B}}_t \mathbf{u}_t\right). \tag{14}$$

The memory adaptor performs a weighted aggregation of the current hidden state with selected previous hidden states, formulated as follows:

$$\mathbf{h}_t = \mathbf{h}_t + \sum_{j \in \mathcal{B}} w_j \mathbf{h}_j \tag{15}$$

where $\mathcal{B}$ is the set of selected indices of forgotten hidden states, and w_j is the learned weights. The indices $\{pos_i\}_{i=1}^{N}$ of N forgotten states are predicted by a linear layer

$$pos_i = \mathcal{P}\left(\mathbf{h}_t\right), \quad 1 \le i \le N. \tag{16}$$

To enhance the modeling of positive and negative correlations, the weights are computed as

$$Pcof_i = \text{Softmax}\left(\text{ReLU}\left(\mathcal{C}\left(\mathbf{h}_t\right)\right)\right) \tag{17}$$

$$Ncof_i = \text{Softmax}\left(\text{ReLU}\left(\mathcal{C}\left(-\mathbf{h}_t\right)\right)\right). \tag{18}$$

The final memory-enhanced hidden state is

$$\mathbf{h}_t = \mathbf{h}_t + \left(\sum_{i=1}^{N} pcof_i + \sum_{i=1}^{N} ncof_i\right) \mathbf{h}_{pos_i}. \tag{19}$$

2) Spatial Adaptor: Before producing the output, a spatial adaptor $\mathcal{S}(\cdot)$ is used to recover local spatial information potentially lost during sequential processing. The adaptor reshapes the features to 2-D and applies depthwise separable convolution as follows:

$$\mathbf{y}_t = \mathcal{S}\left(\bar{\mathbf{C}}_t \mathbf{h}_t + \mathbf{D}\mathbf{u}_t\right). \tag{20}$$

The adaptor-S6 block augments the S6 block with both memory and spatial adaptors, effectively alleviating long-range forgetting and enhancing spatial continuity, thereby improving the model's ability to capture complex spatial–temporal dependencies in multimodal RS data.

E. Multimodal Scale-Aware Fusion Module

As illustrated in Fig. 4, the MSAFM serves as a pivotal component in AutoSAM for capturing local detail information across modalities. MSAFM draws inspiration from the concept of dynamic convolution (DConv) [50], [51], enabling adaptive multiscale feature extraction and fusion. Given the multimodal input features $\mathbf{X}_k$ and $\mathbf{X}_{\Delta\backslash k}$, we first generate two types of intermediate representations: the concatenated feature $\mathbf{Z}_{\text{cat}}$ and the additive feature $\mathbf{Z}_{\text{add}}$. These are calculated as follows:

$$\mathbf{Z}_{\text{cat}} = \text{ReLU}\left(\text{Cat}\left(\mathbf{X}_k, \mathbf{X}_{\Delta\backslash k}\right)\right) \tag{21}$$

$$\mathbf{Z}_{\text{add}} = \text{ReLU}\left(\text{Add}\left(\mathbf{X}_k, \mathbf{X}_{\Delta\backslash k}\right)\right). \tag{22}$$

Next, following the referenced DConv approach, we compute dynamic scale attributes for each convolution kernel K_i along four dimensions: spatial (α_i^s), input channel (α_i^c), output channel (α_i^o), and kernel index (α_i^e). These attention weights adaptively enhance informative scale components while suppressing ineffective ones. The scale-aware fusion output for each representation $T \in \{\text{Cat}, \text{Add}\}$ is as follows:

$$\mathbf{X}_{\text{fuse}}^{T} = \sum_{i=1}^{n} \left(K_i \otimes \alpha_i^s \otimes \alpha_i^c \otimes \alpha_i^o \otimes \alpha_i^e\right) \otimes \mathbf{Z}_T \tag{23}$$

where $\otimes$ denotes appropriate element-wise or channel-wise multiplication. To achieve comprehensive and balanced scale fusion, two learnable coefficients α and β are initialized (typically $\alpha = \beta = 0.5$) to dynamically weigh the contributions of the additive and concatenated scale-fused features. The final fused feature is given by

$$\mathbf{X}_{\text{fuse}} = \alpha\, \mathbf{X}_{\text{fuse}}^{\text{Add}} + \beta\, \mathbf{X}_{\text{fuse}}^{\text{Cat}} \tag{24}$$

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where $\mathbf{X}_{\text{fuse}}$ denotes the output of the MSAFM, which aggregates and balances multiscale information from both modalities in an adaptive manner. This module leverages dynamic convolutional kernels and learnable attention to capture effective multiscale contextual dependencies, providing a robust and flexible mechanism for multimodal feature fusion.

F. Mixture-of-Class-Expert Auto-Prompting Module

The MMoEAPM is the core component responsible for the model's self-prompting capability. As shown in Fig. 2(c), this module integrates dictionary learning with an MoEs architecture to enable adaptive class-aware feature prompting. Specifically, for each of the two MoE branches, we initialize $2 \times num_cls$ learnable tensor dictionaries to encode multimodal class information, comprising `num_cls` class-related dictionaries $\mathbf{D}_k$ and `num_cls` class-unrelated dictionaries. The class-related dictionaries facilitate the selection of class-specific information for pseudo-label generation, while the unrelated dictionaries focus on learning class-agnostic representations during each prediction. For each modality k, to preemptively capture the characteristic distribution of class tensors, we activate `num_cls` class tensors in $\mathbf{D}_k$ via the modality-specific gating network $\mathbf{G}_k(\cdot)$. These activated class tensors are then modulated by the corresponding multimodal activation features and aggregated, which enhances salient class-relevant representation and suppresses irrelevant responses. The fused class feature is calculated as

$$\mathbf{Z}_{\Delta}^{c} = \sum_{cls=1}^{num_cls} \left(\mathbf{G}_k\left(f_k\right)\mathbf{D}_k^{cls} + \mathbf{G}_{\Delta\backslash k}\left(f_{\Delta\backslash k}\right)\mathbf{D}_{\Delta\backslash k}^{cls}\right). \tag{25}$$

Inspired by the decoder design in Sementer [52], the multimodal class tensors are first summed and concatenated with $\mathbf{Z}_{\Delta}^{c}$, followed by two layers of transformer blocks $\varphi_{\text{TB}}(\cdot)$ for improved class-feature interaction. After processing, the two parts are separated by a Split operation

$$\begin{aligned}\mathbf{D}_{\Delta}^{c} = &\ \text{Cat}\left(\left[\mathbf{D}_k^1, \ldots, \mathbf{D}_k^{num_cls}\right]\right) \\ &+ \ \text{Cat}\left(\left[\mathbf{D}_{\Delta\backslash k}^1, \ldots, \mathbf{D}_{\Delta\backslash k}^{num_cls}\right]\right) \qquad (26)\\ \mathbf{Z}_{\Delta}^{c}, \mathbf{D}_{\Delta}^{c} = &\ \text{Split}\left(\varphi_{\text{TB}}\left(\text{Cat}\left(\mathbf{Z}_{\Delta}^{c}, \mathbf{D}_{\Delta}^{c}\right)\right)\right). \qquad (27)\end{aligned}$$

The final pseudo-mask $\mathbf{p}_{\mathcal{GT}}$ is then generated via a self-attention mechanism and upsampling

$$\mathbf{p}_{\mathcal{GT}} = \text{Up}\left(\text{Softmax}\left(\frac{\mathbf{D}_{\Delta}^{c}\left(\mathbf{Z}_{\Delta}^{c}\right)^{\top}}{\sqrt{d}}\right)\mathbf{Z}_{\Delta}^{c}\right). \tag{28}$$

To further optimize the self-prompting process, we adopt a segmentation-based self-distillation strategy. Specifically, we compute the KL divergence between the generated pseudo labels $\mathbf{p}_{\mathcal{GT}}$ and the output $\mathbf{P}_{\mathcal{GT}}$ of the mask decoder

$$\mathcal{L}_{SD} = \varphi_{\text{KL}}\left(\mathbf{p}_{\mathcal{GT}}, \mathbf{P}_{\mathcal{GT}}\right) \tag{29}$$

where $\varphi_{\text{KL}}(\cdot)$ denotes the KL divergence, providing more accurate guidance for subsequent predictions. For downstream prompt generation, the mask prompt $\mathbf{pmask}_{cls}^{(i,j)}$ for class cls is obtained by setting the pixel at position (i, j) to 1 if its predicted class in $\mathbf{p}_{\mathcal{GT}}$ equals cls, and 0 otherwise

$$\mathbf{pmask}_{cls}^{(i,j)} = \begin{cases}1, & \text{if } p_{gt}^{(i,j)} = cls \\ 0, & \text{otherwise.}\end{cases} \tag{30}$$

The point prompt for class cls, $\mathbf{point}_{cls}$, is sampled randomly from locations with $\mathbf{pmask}_{cls}^{(i,j)} = 1$

$$\mathbf{point}_{cls} = \text{RandomP}\left(\mathbf{pmask}_{cls}^{(i,j)} = 1\right). \tag{31}$$

The bounding box prompt for class cls, $\mathbf{box}_{cls}$, is defined by the top-left and bottom-right coordinates among all pixels with value 1 in $\mathbf{pmask}_{cls}^{(i,j)}$

$$\mathbf{box}_{cls} = \left[\ \varphi_{\text{LT}}\left(\mathbf{pmask}_{cls}\right),\ \varphi_{\text{RB}}\left(\mathbf{pmask}_{cls}\right)\ \right] \tag{32}$$

where $\varphi_{\text{LT}}(\cdot)$ returns the smallest (i, j) coordinates among all foreground pixels (value 1), corresponding to the top-left corner; $\varphi_{\text{RB}}(\cdot)$ returns the largest (i, j) coordinates, corresponding to the bottom-right corner. Different from existing prompt-based methods that mainly rely on manual or fixed prompts, our MMoEAPM enables dynamic, category-aware auto-prompt generation through an MoEs and dictionary learning scheme, which significantly enhances the autonomy and adaptability of prompt-driven segmentation.

G. Loss Function

The overall training objective comprises three components: the cross-entropy loss ($\mathcal{L}_{\text{ce}}$), the dice loss ($\mathcal{L}_{\text{dice}}$), and the Kullback–Leibler (KL) divergence loss ($\mathcal{L}_{\text{kl}}$). Specifically, $\mathcal{L}_{\text{ce}}$ and $\mathcal{L}_{\text{dice}}$ are employed to supervise both the pseudo labels and the predictions of the mask decoder, while $\mathcal{L}_{\text{kl}}$ regularizes the consistency between the pseudo labels and the predicted segmentation masks, thus facilitating effective self-distillation. The total loss function is formulated as

$$\mathcal{L}_{\text{all}} = \mathcal{L}_{\text{dice}} + \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{kl}}. \tag{33}$$

The dice loss is widely adopted in image segmentation, particularly for scenarios with pronounced class imbalance. Let p_i be the predicted probability for pixel i belonging to the foreground, y_i the corresponding ground-truth label, and N the total number of pixels. The dice coefficient is defined as

$$\text{Dice} = \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i}. \tag{34}$$

Accordingly, the dice loss is given by

$$\mathcal{L}_{\text{dice}} = 1 - \text{Dice}. \tag{35}$$

The pixel-wise binary cross-entropy loss quantifies the discrepancy between the predicted probability distribution and the ground truth, and is formulated as

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1 - y_i\right)\log\left(1 - p_i\right)\right]. \tag{36}$$

While effective for per-pixel classification, cross-entropy loss is sensitive to class imbalance and may be dominated by background pixels. The KL divergence loss measures the divergence between two probability distributions. In our

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framework, it is utilized to align the model's predicted distribution p_i with the pseudo label distribution y_i, encouraging knowledge transfer and regularization via self-distillation. The symmetric form for binary segmentation is as follows:

$$\mathcal{L}_{\text{kl}} = \frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\frac{y_i}{p_i} + (1-y_i)\log\frac{1-y_i}{1-p_i}\right]. \tag{37}$$

This symmetric formulation, which is closely related to the Jensen–Shannon divergence, is often adopted for improved numerical stability. Alternatively, the standard (asymmetric) KL divergence, $\sum_i y_i \log y_i/p_i$, can be applied depending on the specific distillation direction. The combination of $\mathcal{L}_{\text{dice}}$, $\mathcal{L}_{\text{ce}}$, and $\mathcal{L}_{\text{kl}}$ enables the model to jointly optimize segmentation accuracy, class balance, and knowledge transfer, thereby enhancing both the robustness and generalization capability.

IV. Experiments and Analysis

A. Dataset Description

1) Dongying Dataset [53]: The Dongying dataset is designed for high-resolution land cover classification, comprising 7831 pairs of optical and SAR images, each with a spatial resolution of 1 m and a size of 256×256 pixels. The data were collected over Dongying City, Shandong Province, China, with optical images from the Gaofen-2 satellite and SAR images from Gaofen-3. The dataset covers five land cover categories: buildings, water bodies, farmland, vegetation, and roads, with pixel-level annotations. Following the original protocol, we split the dataset equally into training and testing sets. Notably, the labels include many unannotated pixels (value 255). For training convenience, these are reassigned to a value of 5, but this class is excluded from all evaluation metrics.

2) VALID Dataset [54]: The VALID dataset is constructed from synthetic urban scenes rendered in Unreal Engine 4, encompassing six representative environments such as airports, city centers, villa districts, European streets, coastal towns, and snowy mountains. The dataset simulates diverse weather and lighting conditions, including sunny, dusk, night, snow, and fog. Images are captured at 20, 50, and 100 m altitude with a resolution of 1024 × 1024 pixels and pixel-level annotations for 31 semantic categories. Following the original split, half of the 6690 Opt-DSM image pairs are used for training, one-sixth for validation, and one-third for testing.

3) RS-RGB-T Dataset [55]: The RS-RGB-T dataset consists of 4024 pairs of aligned RGB and thermal images of urban roads, acquired by UAV-mounted cameras. Of these, 2514 pairs are from daytime and 1510 from nighttime. Each image has a resolution of 640 × 512 pixels and is annotated at the pixel level across nine classes: unlabeled, road, building, motorcycle, car, truck, tree, human, and traffic facilities. Following the protocol in [55], the dataset is split into training, validation, and test sets in a 7:1:2 ratio, maintaining the same day/night distribution in each subset.

4) BJRoad Dataset [56]: The BJRoad dataset provides paired optical and trajectories modality data, the latter derived from 50 million GPS traces collected from 28 000 vehicles. These traces are aggregated to form trajectory images. The dataset contains 1350 optical–trajectories image pairs, each covering roughly 100 km^2 with a resolution of 1024 × 1024 pixels. For training, images are cropped into 256×256 patches, generating 21 600 pairs in total, which are divided into 70% for training, 10% for validation, and 20% for testing. The segmentation task involves two classes: road and others.

B. Evaluation Metrics

To quantitatively evaluate the multiclass segmentation performance of AutoSAM, we adopt three widely used metrics: overall accuracy (OA), mean $F1$-score (mF1), and mean intersection over union (mIoU).

OA measures the proportion of correctly classified pixels among all pixels

$$\text{OA} = \frac{\sum_{p=1}^{P}\mathbb{I}\left(\hat{y}_p = y_p\right)}{P} \tag{38}$$

where P is the total number of pixels, $\hat{y}_p$ is the predicted class, y_p is the ground-truth class, and $\mathbb{I}(\cdot)$ is the indicator function.

mF1 evaluates the balance between precision and recall for each class c

$$F1_c = \frac{2 \times TP_c}{2 \times TP_c + FP_c + FN_c} \tag{39}$$

where TP_c, FP_c, and FN_c denote the number of true positives, false positives, and false negatives for class c, respectively. The mF1 is then

$$\text{mF1} = \frac{1}{K}\sum_{c=1}^{K} F1_c \tag{40}$$

with K being the total number of classes.

mIoU quantifies the overlap between the predicted and ground-truth regions

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c}. \tag{41}$$

The final mIoU is the average across all classes

$$\text{mIoU} = \frac{1}{K}\sum_{c=1}^{K}\text{IoU}_c. \tag{42}$$

These metrics provide a comprehensive assessment of pixel-level accuracy, class-wise balance, and segmentation quality.

C. Implement Details

All experiments are conducted on a workstation equipped with an Intel Xeon Gold 6133 CPU (2.50 GHz), 24-GB RAM, an NVIDIA GeForce RTX 3090 GPU, and Ubuntu 18.04. The implementation is based on PyTorch 2.0.0. Model optimization is performed using the AdamW optimizer with an initial learning rate of 0.0001 and a weight decay of 0.01. The learning rate is kept constant throughout the 50 training epochs. Due to GPU memory limitations, the batch size is set to 2 for all training runs. During training, standard data augmentation strategies, including random cropping and horizontal flipping, are applied to improve generalization. All other hyperparameters remain at their default values unless otherwise specified. For involving the SAM-based structure, we use the official SAM checkpoint "`sam_vit_b.pth`" as the model initialization. This ensures that our implementation and code are aligned with the reported experimental results.

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TABLE I

QUANTITATIVE COMPARISON OF AUTOSAM AND 16 STATE-OF-THE-ART SEMANTIC SEGMENTATION METHODS ON THE RS-RGB-T DATASET. THE TABLE REPORTS THE IOU (%) FOR NINE FOREGROUND CLASSES AND THREE OVERALL PERFORMANCE METRICS: MIOU, MF1, AND OA. THE HIGHEST VALUES ARE MARKED IN RED, AND THE SECOND-BEST IN BLUE

Type	Method	Unlabeled	Road	Building	Motorcycle	Car	Truck	Tree	Human	Traffic	mF1	OA	mIoU
UniModal	UNetFormer	68.65	84.58	82.04	69.05	82.11	84.95	78.07	44.82	56.73	78.35	83.32	72.33
	Segformer	70.28	85.12	82.68	69.60	82.66	85.50	78.72	45.16	57.10	79.57	84.10	72.98
	TransUNet	70.06	83.25	80.70	67.50	80.60	83.40	76.60	44.02	56.47	77.76	82.17	71.44
	DeepLabv3+	64.55	81.33	78.78	65.61	78.71	81.51	74.52	43.15	55.45	76.12	81.02	69.29
Multimodal	RingMo-SAM	77.01	87.59	86.31	71.60	84.10	86.08	82.16	45.41	60.18	82.98	86.22	75.66
	DDHRNet	68.99	85.34	82.91	69.83	82.89	85.73	78.96	45.29	56.97	79.48	82.16	72.99
	VFuseNet	73.38	85.04	83.59	69.64	83.22	85.14	78.57	43.30	57.10	82.36	84.40	73.22
	FuseNet	74.36	86.01	84.56	70.61	84.19	86.11	79.54	44.27	58.06	83.22	85.07	74.19
	MCANet	74.14	85.80	84.35	70.40	83.98	85.90	79.33	44.06	57.86	81.22	84.22	73.98
	CMFNet	74.29	85.94	84.49	70.54	84.12	86.04	79.47	44.20	57.99	80.10	85.06	74.12
	MambaDiffusion	74.78	86.43	84.98	71.03	84.61	86.53	79.96	44.69	58.48	82.12	85.62	74.61
	MultiSenseSeg	76.83	87.44	86.16	71.48	86.94	87.69	82.01	45.35	60.06	79.27	86.93	75.54
	FTransUNet	75.20	88.31	86.79	72.18	84.62	86.48	79.92	43.23	58.99	83.04	87.49	75.08
	MMFormer	69.36	85.47	83.17	70.15	83.13	85.96	79.34	45.43	57.35	80.07	83.12	73.26
	I^2HN	77.17	87.19	84.58	73.64	85.98	88.80	81.26	45.32	60.21	84.47	88.21	76.01
	CFFormer	79.44	86.95	85.33	71.37	85.51	87.21	80.15	46.11	60.07	85.03	87.49	75.79
	AutoSAM	80.33	89.72	89.89	77.07	89.13	89.74	85.04	49.17	64.33	87.29	91.06	79.38

D. Comparison With State-of-the-Art Methods

To demonstrate the effectiveness of the proposed AutoSAM model, we conduct a comprehensive comparison with 16 state-of-the-art semantic segmentation methods, including UNetFormer [57], Segformer [11], TransUNet [58], Deeplabv3 + [59], CFFormer [60], DDHRNet [53], VFuseNet [61], FuseNet [62], MCANet [63], CMFNet [64], MambaDiffusion [65], MultiSenseSeg [66], FTransUNet [67], MMFormer [68], I^2HN [69], and RingMo-SAM [37]. All baseline methods were retrained and evaluated under identical experimental settings, including the same training datasets, input resolutions, data augmentations, optimizer configurations, and hardware environment as AutoSAM. For models that support automatic prompting (e.g., MultiSenseSeg, MMFormer, and RingMo-SAM in auto-prompt mode), we employed their respective self-prompting mechanisms. For models requiring manual prompts (such as SAM and RingMo-SAM in manual mode), we followed the original protocol and adopted standard point-based or bounding-box prompts. The evaluation is conducted on four benchmark datasets. In Sections IV-D1–IV-D4, we present detailed results and analyses of both the comparative experiments and ablation studies.

1) Results on RS-RGB-T Dataset: Table I presents a comprehensive quantitative comparison between AutoSAM and 16 state-of-the-art semantic segmentation methods on the RS-RGB-T dataset. AutoSAM achieves the best performance across all metrics, with an mIoU of 79.38%, OA of 91.06%, and mF1 of 87.29%, outperforming the second-best method by 3.37%, 2.85%, and 2.26%, respectively. Notably, AutoSAM attains the highest IoU in every category, with particularly significant improvements in challenging small-object classes such as human and motorcycle, exceeding the runner-up by 3.06% and 3.43%. These results demonstrate that AutoSAM excels at handling multimodal semantic segmentation, benefiting from both the strong visual representation capability of the SAM backbone and the effective transfer of knowledge from natural to RS domains. In addition, the integration of the MSAFM module enables the model to capture fine-grained features, further enhancing segmentation performance across objects of various scales. Moreover, the overall results confirm that multimodal methods generally outperform unimodal approaches. The complementary thermal (TIR) modality provides crucial information under challenging conditions, such as nighttime, effectively compensating for the limitations of optical imagery. Fig. 5 visualizes segmentation results for AutoSAM and competing methods on two representative scenes from the RS-RGB-T dataset. These scenarios highlight cases where either the optical or thermal modality is degraded or missing, such as a lack of thermal detail in the left scene and the absence of optical cues in the nighttime right scene. AutoSAM demonstrates robust segmentation under both conditions, owing to its comprehensive multimodal feature fusion and contextual modeling. For example, in the right scene, AutoSAM accurately distinguishes the tree class from the background, where most other methods fail. In the left scene, it precisely delineates small objects like a motorcycle, further illustrating its superior capacity for fine-grained segmentation. These results underscore the robustness and effectiveness of AutoSAM in complex real-world multimodal scenarios.

2) Results on Dongying Dataset: Table II presents the quantitative segmentation results of AutoSAM and 16 comparative methods on the Dongying (Optical-SAR) dataset. AutoSAM achieves state-of-the-art performance, with an mIoU of 93.98%, OA of 97.67%, and mF1 of 96.07%, outperforming the second-best method (CFFormer) by 2.16%, 2.30%, and 0.95%, respectively. Across individual categories, AutoSAM ranks first in IoU for water, farmland, greenery, and road, and second for building, where the SAR modality poses significant challenges due to weak boundary contrast. Despite this, our approach maintains highly competitive results for building, highlighting the strong generalization ability of AutoSAM in diverse multimodal RS scenarios. These results underscore the adaptability of AutoSAM to different modality combinations. The superior performance is attributed to the

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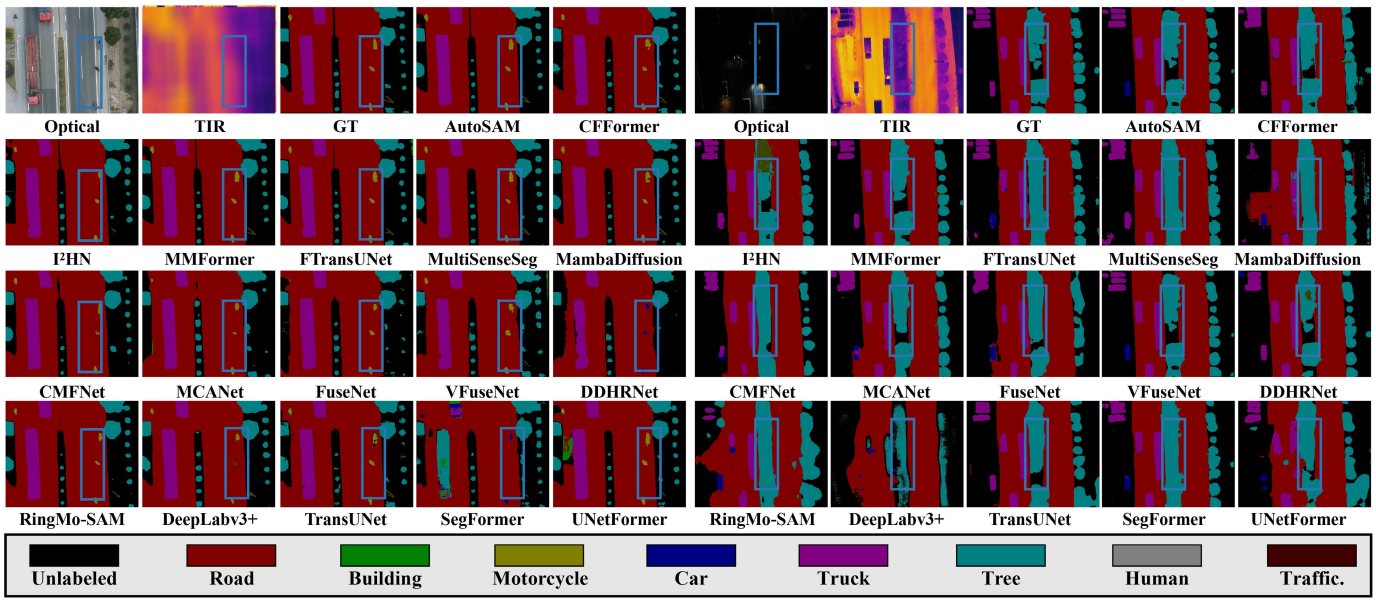


Fig. 5. Qualitative comparison of segmentation results on the RS-RGB-T dataset. Predictions from AutoSAM and 16 methods are shown for two scenes with missing or degraded modal information. AutoSAM demonstrates superior accuracy and robustness, particularly for small or ambiguous objects.

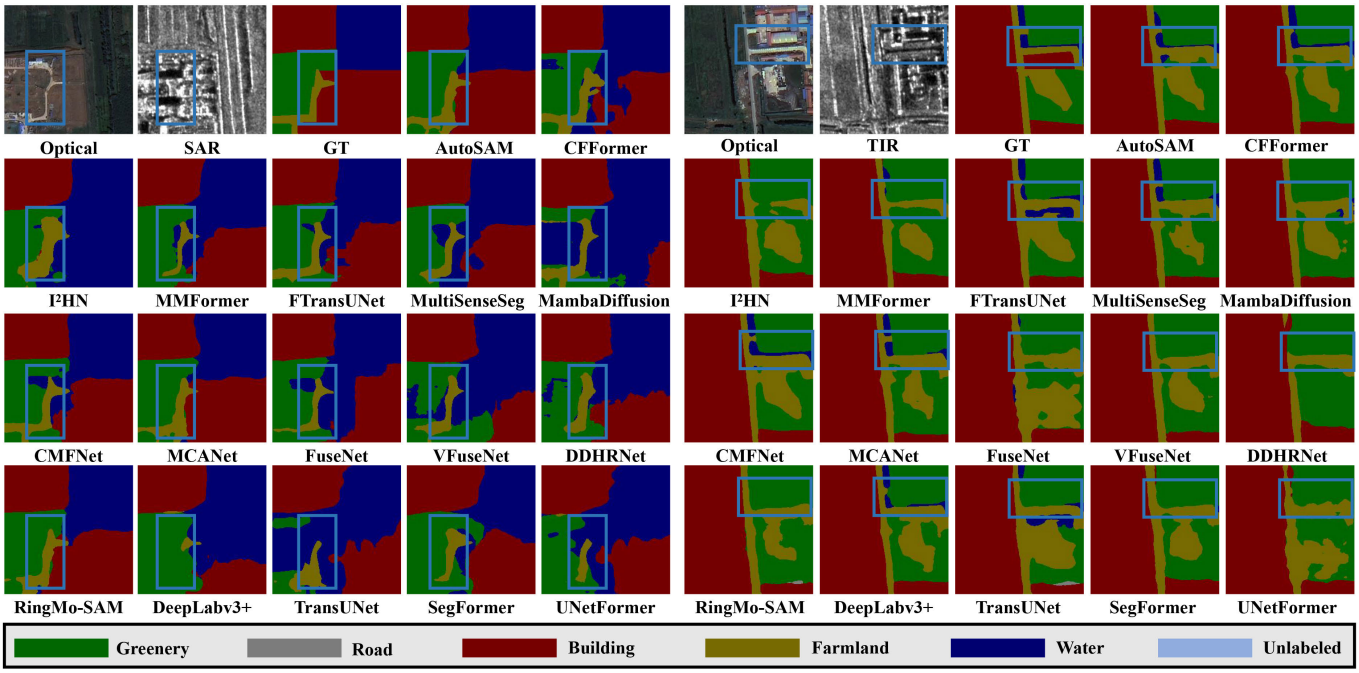


Fig. 6. Qualitative comparison of segmentation results on the Dongying dataset. Predictions from AutoSAM and 16 representative methods are shown for two scenes. AutoSAM demonstrates superior accuracy and detail preservation, especially in complex or ambiguous regions (highlighted in blue).

effective fusion of optical and SAR features, leveraging the strengths of each modality and the foundational SAM model's robust representation. Notably, our model demonstrates consistent improvements over both unimodal and other multimodal competitors, validating its modality generalization capability. Fig. 6 provides qualitative comparisons on two representative scenes from the Dongying dataset. The visualizations show that AutoSAM delivers the most accurate and coherent segmentation results, particularly within the blue-highlighted regions. AutoSAM excels at both capturing fine-grained structural details and modeling global context, outperforming other methods in areas with complex boundaries or ambiguous features. These findings verify the effectiveness and robustness of AutoSAM in challenging multimodal segmentation tasks.

3) Results on VALID Dataset: Table III presents the quantitative comparison of AutoSAM with 16 state-of-the-art methods on the VALID dataset, which involves the challenging Opt-DSM modality and contains over thirty categories. Different from other datasets, VALID suffers from severe modality information deficiency and category imbalance,

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TABLE II

QUANTITATIVE COMPARISON OF AUTOSAM AND 16 STATE-OF-THE-ART SEGMENTATION METHODS ON THE DONGYING DATASET. IOU (%) FOR FIVE FOREGROUND CLASSES AND THREE OVERALL METRICS (MIOU, MF1, AND OA) ARE REPORTED. THE BEST RESULTS ARE SHOWN IN RED, AND THE SECOND-BEST IN BLUE

Type	Method	Building	Water	Farmland	Greenery	Road	mF1	OA	mIoU
UniModal	UNetFormer	90.14	87.28	88.32	90.06	89.55	92.81	93.31	89.07
	Segformer	87.22	86.28	90.08	91.02	89.30	92.52	92.64	88.78
	TransUNet	89.29	85.46	91.43	90.44	89.43	93.01	93.26	88.81
	DeepLabv3+	85.13	87.29	91.80	90.51	86.42	92.52	93.01	88.23
Multimodal	RingMo-SAM	93.68	87.60	88.81	90.80	94.66	94.52	94.89	91.11
	DDHRNet	89.68	88.34	90.66	91.30	86.06	93.23	93.92	89.21
	V_FuseNet	90.50	86.82	88.62	91.66	92.11	93.54	93.78	89.94
	FuseNet	90.10	86.84	89.98	91.94	93.54	93.22	95.04	90.48
	MCANet	89.08	88.66	89.94	91.31	91.21	93.92	94.77	90.04
	CMFNet	90.63	87.37	89.31	91.62	90.17	93.10	94.12	89.82
	MambaDiffusion	92.30	89.55	92.70	89.76	91.29	94.13	94.52	91.12
	MultiSenseSeg	90.24	87.88	91.46	91.11	90.71	94.13	94.72	90.28
	FTransUNet	92.31	87.57	90.74	92.54	92.04	94.62	94.77	91.04
	MMFormer	89.01	87.82	91.74	92.81	91.77	93.71	94.59	90.63
	I^2HN	92.49	89.13	91.83	91.79	91.87	92.89	95.02	91.42
	CFFormer	94.60	88.60	89.03	93.62	93.25	95.12	95.37	91.82
	AutoSAM	93.22	91.89	93.46	95.30	96.02	96.07	97.67	93.98

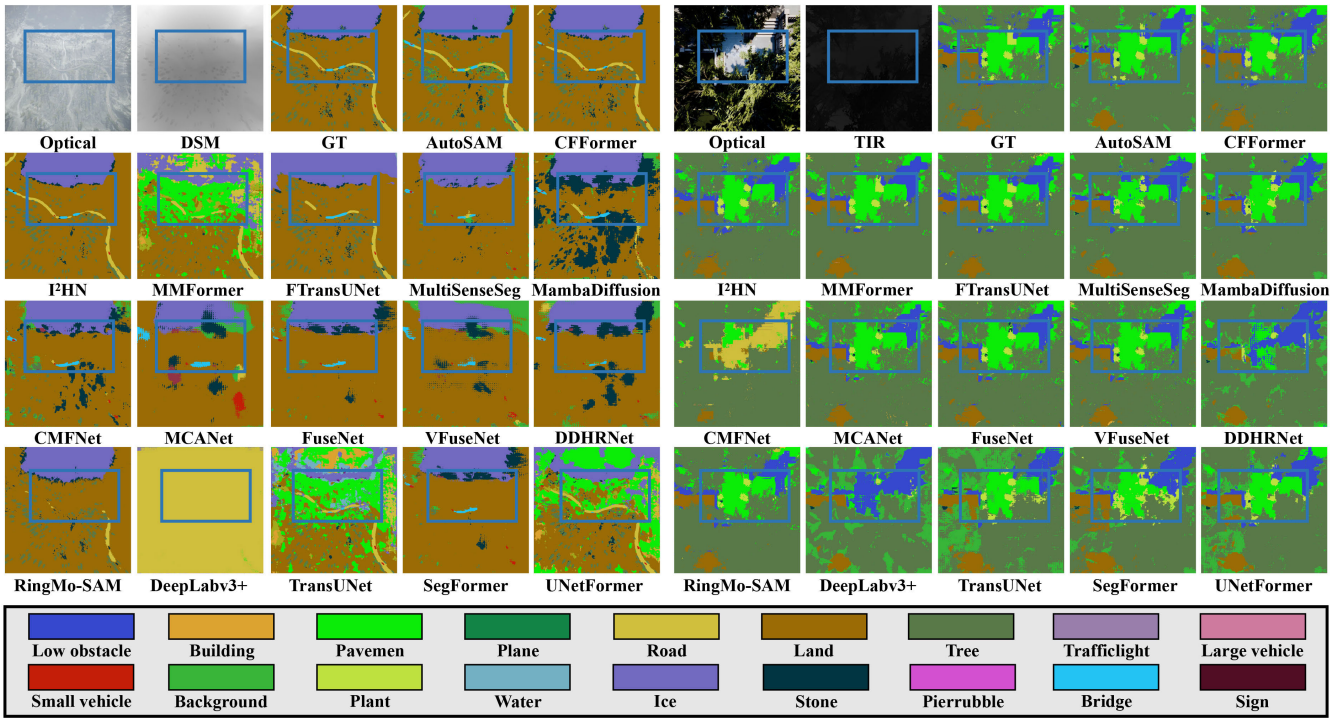


Fig. 7. Qualitative comparison of semantic segmentation results on the VALID dataset. Predictions from AutoSAM and 16 representative methods are shown for two challenging scenes. AutoSAM demonstrates superior accuracy and robustness, especially in scenes with degraded modality information and complex category boundaries (highlighted in blue).

increasing the difficulty of semantic segmentation. Despite this, AutoSAM achieves the best overall performance, with mIoU, OA, and mF1 scores of 75.64%, 90.62%, and 84.79%, respectively, outperforming the second-best method by a clear margin. Due to space constraints, per-category results are not listed; nonetheless, AutoSAM consistently surpasses both unimodal and other multimodal baselines, demonstrating strong robustness and generalization. The superior performance of AutoSAM can be attributed to the integration of a scale-aware strategy and the MMoEAPM self-prompting mechanism, which enables accurate object localization even under adverse conditions.

In particular, the self-prompting module allows the model to dynamically identify and segment targets in scenes with limited optical or DSM information, effectively mitigating the challenges posed by missing or ambiguous data. Fig. 7 visualizes the segmentation results of AutoSAM and competing methods on two representative scenes. In the left scene, both optical and DSM modalities are degraded due to weather, resulting in many methods misclassifying categories such as

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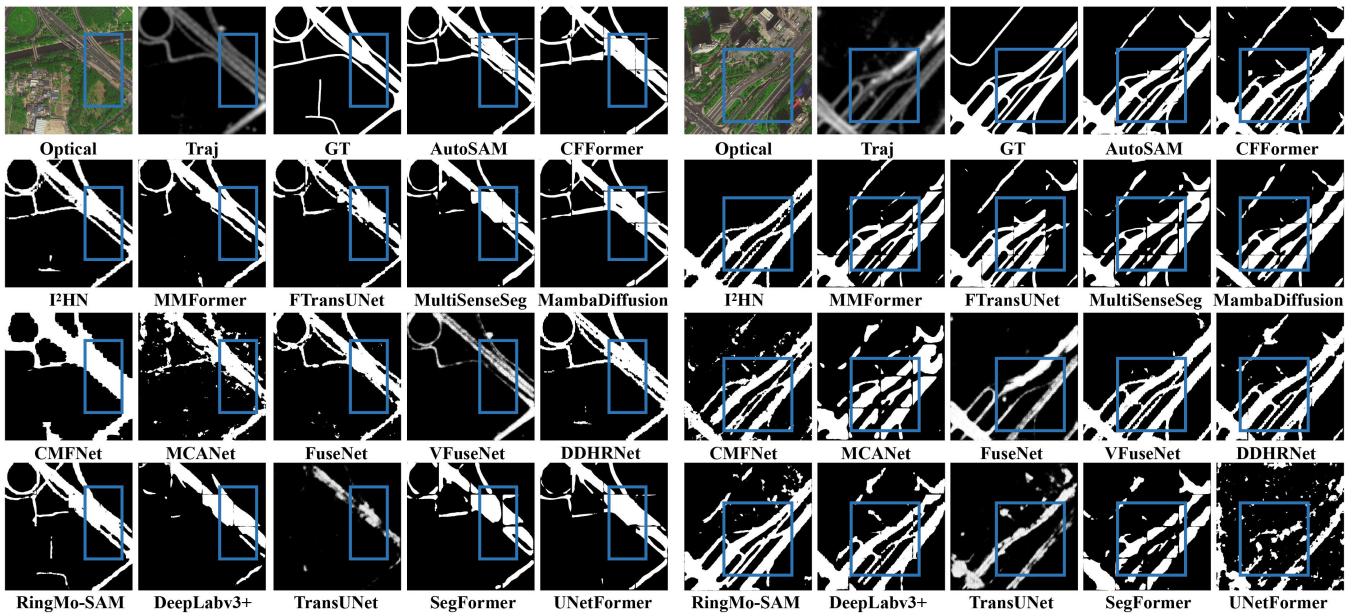


Fig. 8. Qualitative comparison of segmentation results on the BJRoad dataset. Predictions from AutoSAM and 16 representative methods are shown for two scenes. AutoSAM demonstrates superior accuracy and continuity in road segmentation, especially under complex backgrounds or trajectory data sparsity.

TABLE III

QUANTITATIVE COMPARISON OF AUTOSAM AND 16 STATE-OF-THE-ART METHODS ON THE VALID DATASET. MF1, OA, AND MIOU ARE REPORTED AS OVERALL METRICS. THE BEST RESULTS ARE SHOWN IN RED, AND THE SECOND-BEST IN BLUE

Type	Method	mF1	OA	mIoU
UniModal	UNetFormer	68.81	73.31	55.07
	Segformer	67.92	71.57	55.28
	TransUNet	68.34	72.13	54.30
	Deeplabv3+	67.05	70.01	53.16
Multimodal	RingMo-SAM	81.03	85.96	67.36
	DDHRNet	71.44	76.37	57.61
	V_FuseNet	71.17	75.18	56.29
	FuseNet	75.16	79.04	58.74
	MCANet	74.92	77.97	56.14
	CMFNet	77.11	80.01	59.72
	MambaDiffusion	80.84	85.35	65.56
	MultiSenseSeg	83.01	88.10	69.10
	FTransUNet	80.34	86.73	67.45
	MMFormer	81.71	86.39	66.31
	I^2HN	83.19	87.36	71.12
	CFFormer	81.23	84.12	65.38
	AutoSAM	84.79	90.62	75.64

TABLE IV

QUANTITATIVE COMPARISON OF AUTOSAM AND 16 STATE-OF-THE-ART METHODS ON THE BJROAD DATASET. MF1, OA, AND MIOU ARE REPORTED AS OVERALL METRICS. THE BEST RESULTS ARE SHOWN IN RED, AND THE SECOND-BEST IN BLUE

Type	Method	mF1	OA	mIoU
UniModal	UNetFormer	74.25	76.54	61.33
	Segformer	72.39	76.02	60.76
	TransUNet	69.91	74.43	58.07
	Deeplabv3+	70.49	75.82	59.35
Multimodal	RingMo-SAM	76.98	83.83	67.18
	DDHRNet	75.44	81.79	65.08
	V_FuseNet	74.30	80.20	64.16
	FuseNet	74.00	79.04	63.23
	MCANet	74.72	78.92	64.06
	CMFNet	74.19	78.23	63.04
	MambaDiffusion	73.91	77.83	63.89
	MultiSenseSeg	73.54	81.19	65.03
	FTransUNet	75.94	82.02	66.21
	MMFormer	72.33	78.10	63.10
	I^2HN	75.39	81.12	65.96
	CFFormer	76.55	82.31	66.74
	AutoSAM	79.08	85.02	69.48

land and pavement. In contrast, AutoSAM accurately distinguishes these classes and exhibits high precision for small objects like roads and bridges. In the right scene, despite the DSM image providing limited height information, AutoSAM can still delineate the boundaries and locations of complex categories such as trees, pavement, and low obstacles. These qualitative results validate the robustness and effectiveness of AutoSAM in large-scale and weak-modality environments.

4) Results on BJRoad Dataset: Table IV summarizes the quantitative segmentation results of AutoSAM and 16 comparative methods on the BJRoad dataset, which focuses on the challenging Opt-Traj modality. The Traj modality, representing vehicle trajectories, often suffers from data sparsity, particularly on roads with little or no vehicle presence, resulting in severe modality mismatches with optical images. Despite these challenges, AutoSAM achieves state-of-the-art performance, with mIoU, OA, and mF1 scores of 69.48%, 85.02%, and 79.08%, respectively, surpassing the second-best method by 3.52%, 3.83%, and 3.69%. These results demonstrate the strong adaptability and generalization capability of AutoSAM for heterogeneous multimodal data. The superior performance can be attributed to the robust feature representation and cross-modal fusion strategies within AutoSAM, which effectively reconcile the semantic inconsistencies between optical and trajectory data. This enables accurate identification and segmentation of road regions, even in cases where one

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TABLE V

COMPUTATIONAL COMPLEXITY COMPARISON OF AUTOSAM AND REPRESENTATIVE SEGMENTATION MODELS ON THE RS-RGB-T DATASET. FLOPS, PARAMETER COUNT, MEMORY USAGE, AND INFERENCE SPEED (FPS) ARE REPORTED. MIOU VALUES REFLECT SEGMENTATION ACCURACY. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Type	Method	FLOPs (G)	Param. (M)	Memory (MB)	Speed (FPS)	mIoU (%)
UniModal	UNetFormer	**48.3**	**27.4**	1820	57	72.33
	Segformer	41.3	24.7	1610	69	72.98
	TransUNet	102.3	105.1	4260	74	71.44
	Deeplabv3+	68.4	39.1	1900	77	69.29
Multimodal	RingMo-SAM	153.1	120.3	4470	28	75.66
	DDHRNet	70.4	41.2	2270	51	72.99
	V_FuseNet	73.5	48.5	2480	46	73.22
	FuseNet	80.3	47.9	2370	54	74.19
	MCANet	69.2	36.8	2050	65	73.98
	CMFNet	65.1	35.7	1940	63	74.12
	MambaDiffusion	90.3	45.6	2560	47	74.61
	MultiSenseSeg	70.3	42.8	2710	45	75.54
	FTransUNet	87.7	54.3	**2870**	40	75.08
	MMFormer	62.1	49.7	2820	38	73.26
	I^2HN	81.4	38.2	1990	53	76.01
	CFFormer	78.8	44.6	2430	50	75.79
	AutoSAM	75.7	47.9	2740	48	**79.38**

TABLE VI

ABLATION STUDY OF THE PROPOSED MODULES (MMOEAPM, MSAFM, AND CMB) ON FOUR BENCHMARK DATASETS. PERFORMANCE IS REPORTED AS MF1, OA, AND MIOU

Model	MMoEAPM	MSAFM	CMB	Dongying			RS-RGB-T			VALID			BJRoad		
				mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU
AutoSAM	×	×	×	79.85	82.09	80.46	79.04	80.43	69.76	72.69	75.51	66.86	68.30	70.98	59.31
	✓	×	×	91.67	92.37	89.93	84.45	85.92	76.39	81.15	86.21	73.22	76.77	81.02	66.32
	✓	✓	×	93.45	94.11	91.68	85.11	87.57	77.87	82.84	87.69	74.01	77.49	82.10	67.71
	✓	✓	✓	96.07	97.67	93.98	87.29	91.06	79.38	84.79	90.62	75.64	79.08	85.02	69.48

TABLE VII

PARAMETER, FLOPS, GPU MEMORY, AND AVERAGE RUNTIME PER ITERATION FOR EACH ABLATED AUTOSAM VARIANT ON THE VALID DATASET

Model Variant	Parameters (M)	FLOPs (G)	Peak GPU Memory (GB)	Runtime (s/iter)
Baseline	38.2	72.4	14.3	0.41
+MMoEAPM	41.4	74.1	17.7	0.44
+MMoEAPM+MSAFM	43.7	76.5	19.9	0.46
+MMoEAPM+MSAFM+CMB	47.9	75.7	21.7	0.48

modality provides incomplete or misleading cues. Fig. 8 provides qualitative comparisons on two representative BJRoad scenes. In the left example, the optical image contains complex textures, and the river closely resembles the road in both shape and appearance, posing significant confusion for most methods. In the right example, certain road sections lack trajectory information due to the absence of vehicles, challenging the model's ability to maintain spatial continuity. In both scenarios, AutoSAM consistently delivers the most accurate segmentation, precisely distinguishing the slender, discontinuous road segments from complex backgrounds and effectively delineating boundaries in regions of ambiguity. These results further highlight the effectiveness of AutoSAM in real-world, cross-modality road segmentation tasks.

E. Computational Complexity Analysis

Table V presents a comprehensive computational complexity analysis of AutoSAM and several representative segmentation models on the RS-RGB-T dataset, evaluated with 640 × 512 input images on a single NVIDIA RTX A100 GPU. The comparison covers four key aspects: FLOPs, parameter count, memory usage, and inference speed (FPS). AutoSAM demonstrates a balanced tradeoff between efficiency and accuracy. Specifically, it achieves a speed of 48 FPS and 75.7-G FLOPs, outperforming most multimodal competitors such as RingMo-SAM and FTransUNet in real-time inference. Although its speed and FLOPs do not match the fastest models (e.g., CFFormer and MCANet), AutoSAM consistently surpasses them in segmentation accuracy, achieving the highest mIoU of 79.38%. The improved real-time capability is largely attributed to the integration of the linear-complexity VMamba module, replacing the quadratic self-attention mechanism and significantly reducing computational overhead. In terms of model deployment, AutoSAM has 47.9-M parameters and a memory footprint of 2740 MB. While these metrics are not the lowest among the compared methods, they remain within

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TABLE VIII

INDIVIDUAL AND JOINT ABLATION ANALYSIS OF MMOEAPM, MSAFM, AND CMB MODULES ON ALL DATASETS. RESULTS SHOW THE CONTRIBUTION OF EACH MODULE TO SEGMENTATION PERFORMANCE

Model	MMoEAPM	MSAFM	CMB	Dongying			RS-RGB-T			VALID			BJRoad		
				mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU
AutoSAM	✓	×	×	91.67	92.37	89.93	84.45	85.92	76.12	81.15	86.21	73.22	76.77	81.02	66.32
	×	✓	×	83.87	85.93	83.49	77.77	79.18	73.04	74.05	78.15	68.33	72.14	75.09	60.18
	×	×	✓	87.18	89.21	86.59	84.72	86.31	75.97	80.64	85.15	72.85	74.37	77.44	62.94
	✓	✓	✓	96.07	97.67	93.98	87.29	91.06	79.38	84.79	90.62	75.64	79.08	85.02	69.48

TABLE IX

MODULE-LEVEL ABLATION EXPERIMENTS ON FOUR DATASETS. FOR EACH OF THE THREE CORE MODULES IN AUTOSAM (MMOEAPM, CMB, AND MSAFM), WE REPLACE IT WITH TWO STATE-OF-THE-ART ALTERNATIVES, AND REPORT MIOU (%) ON EACH DATASET

Model	Dongying			RS-RGB-T			VALID			BJRoad		
	+MMoEAPM	+CMB	+MSAFM	+MMoEAPM	+CMB	+MSAFM	+MMoEAPM	+CMB	+MSAFM	+MMoEAPM	+CMB	+MSAFM
AutoSAM (Origin)	93.98	93.98	93.98	79.38	79.38	79.38	75.64	75.64	75.64	69.48	69.48	69.48
	+A1	+B1	+C1	+A1	+B1	+C1	+A1	+B1	+C1	+A1	+B1	+C1
AutoSAM (Ablation)	89.83	91.20	90.34	75.21	77.34	77.12	71.40	73.65	71.97	64.37	66.02	66.53
	+A2	+B2	+C2	+A2	+B2	+C2	+A2	+B2	+C2	+A2	+B2	+C2
AutoSAM (Ablation)	90.12	91.32	91.26	74.42	76.13	76.76	72.30	74.00	73.15	65.00	67.19	66.75

a reasonable range for practical deployment. However, there is still room for improvement in model size and memory efficiency, which we identify as a promising direction for future work to explore lightweight visual foundation models tailored for multimodal RS tasks. Overall, AutoSAM strikes an effective balance between computational efficiency and segmentation performance, demonstrating its practical value for real-world multimodal RS applications.

F. Ablation Study of Each Module

To validate the effectiveness of each proposed module in AutoSAM, we conducted comprehensive ablation experiments on four benchmark datasets. In the first ablation (see Table VI), we adopted the prompt-free SAM as the baseline, where the original prompt mechanism is replaced by learnable parameters. We then progressively added MMoEAPM, MSAFM, and CMB modules. When only MMoEAPM is introduced, the model shows a substantial improvement across all datasets, demonstrating that effective self-prompting is essential for accurate segmentation in multimodal scenarios. This is primarily because the SAM architecture heavily relies on prompt cues for object localization, and MMoEAPM successfully provides pseudo-labels that guide the segmentation process. Incorporating MSAFM further enhances the ability to capture fine-grained object details, notably improving segmentation of small or ambiguous targets. Finally, replacing the transformer block with CMB brings further gains, as CMB enables both global context modeling within individual modalities and cross-modal feature interaction, leading to additional performance improvements. To further analyze the scalability of AutoSAM, we report parameter count, FLOPs, peak GPU memory usage, and runtime per iteration for each ablation variant (see Table VII). The second ablation (see Table VIII) investigates the individual and combined effects of each module. We sequentially introduced MMoEAPM, MSAFM, and CMB into the prompt-free SAM, each time keeping the other modules absent. Results indicate that MMoEAPM delivers the most significant performance boost, underscoring the critical role of effective prompting in multimodal segmentation. Both CMB and MSAFM also yield improvements, with CMB demonstrating that VMamba-based blocks can effectively replace transformers for global and cross-modal feature extraction.

To quantify the individual contributions of each core module in AutoSAM, we conducted a systematic module-level ablation study, as summarized in Table IX. For each of the three key components: 1) auto-prompt module; 2) Mamba block; and 3) multimodal fusion scheme, we removed the corresponding module from AutoSAM and substituted it with two state-of-the-art alternatives from recent methods, (A1) adaptive prompt generation [32], (A2) auto-prompt generator [70], (B1) graph Mamba [71], (B2) Natten-Mamba [72], (C1) synergistic high-order interaction paradigm [73], and (C2) spatial-frequency domain fusion [74]. Specifically, each time one module (MMoEAPM, CMB, and MSAFM) is ablated, we incorporated the corresponding alternative within its category and evaluated the resulting models on four benchmark datasets using the mIoU metric. Experimental results in Table IX show that after replacing each of the core modules (MMoEAPM, CMB, and MSAFM) with state-of-the-art alternatives, the mIoU scores on all four datasets exhibit a noticeable decline compared to the original AutoSAM configuration. Specifically, substituting the Auto-Prompt module with either Adaptive Prompt Generation or Auto-Prompt Generator leads to reduced segmentation accuracy, indicating the advantage of MMoEAPM. Similarly, replacing the CMB or MSAFM modules with their

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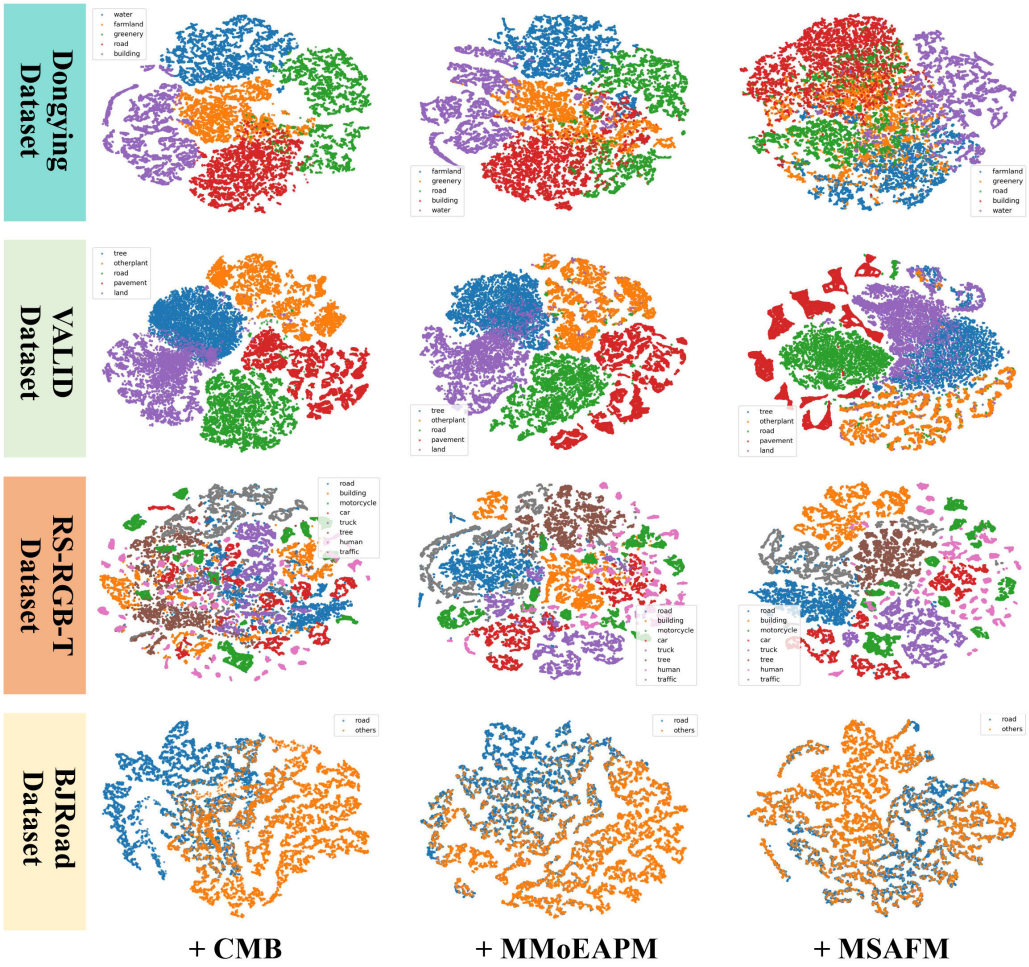


Fig. 9. t-SNE visualization of feature distributions after adding different modules in the ablation study. Progressive module integration leads to more compact and discriminative feature clustering across different datasets.

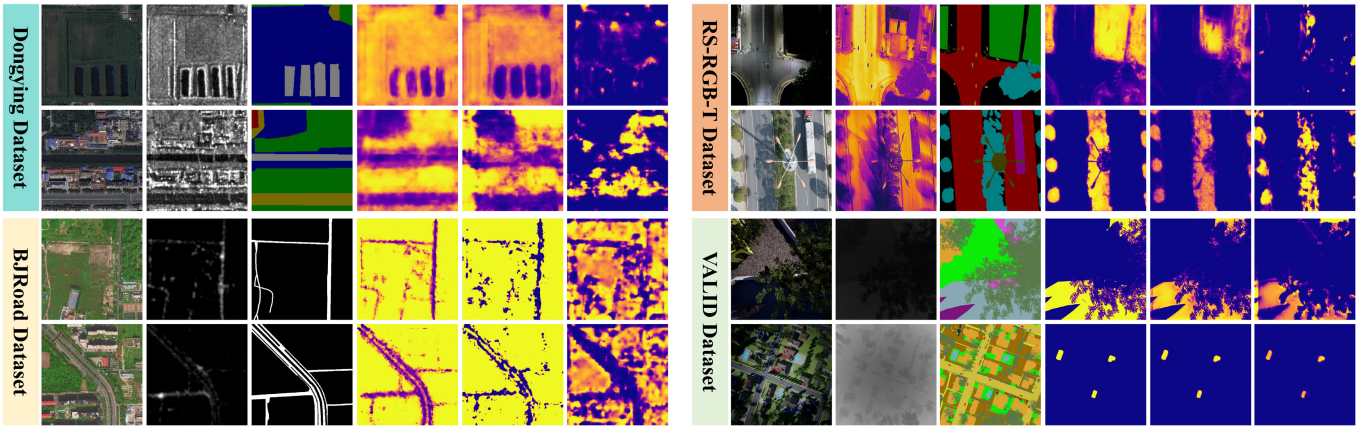


Fig. 10. Visualization of heatmaps produced by different modules in the ablation study. The integration of MMoEAPM, MSAFM, and CMB progressively enhances target localization and boundary definition in segmentation results.

respective alternatives results in lower mIoU, demonstrating the superior effectiveness of our proposed cross-modal modeling and fusion strategy. These findings directly confirm that each module in AutoSAM plays a critical and irreplaceable role in achieving optimal performance on multimodal segmentation tasks.

To further illustrate the benefits of each module, we visualize the t-SNE feature distributions and heatmaps of the intermediate activations in Figs. 9 and 10. The t-SNE plots reveal that the integration of each module leads to more compact and discriminative feature clustering, while the heatmaps highlight the progressive enhancement in localization and

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TABLE X

ABLATION STUDY OF AUTOSAM WITH AND WITHOUT AUXILIARY MODAL INPUTS ON FOUR BENCHMARK DATASETS. RESULTS DEMONSTRATE THE IMPACT OF MULTIMODAL FUSION ON SEGMENTATION PERFORMANCE, AS WELL AS MODEL SCALABILITY IN TERMS OF PARAMETER COUNT, GPU MEMORY, AND RUNTIME PER ITERATION

Model	Optical	Other Modal	Dongying			RS-RGB-T			VALID			BJRoad			Para.	Mem.	Time.
			mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU			
AutoSAM	✓	×	90.19	92.33	91.02	84.41	87.22	75.63	81.05	87.48	72.92	77.00	82.16	67.38	33.5	15.6	0.41
AutoSAM	✓	✓	96.07	97.67	93.98	87.29	91.06	79.38	84.79	90.62	75.64	79.08	85.02	69.48	47.9	21.7	0.48

TABLE XI

COMPARISON OF USER PROMPTS AND SELF-PROMPTS IN MRSAUTOSAM ACROSS FOUR DATASETS. RESULTS HIGHLIGHT THE INFLUENCE OF DIFFERENT PROMPT PARADIGMS ON SEGMENTATION ACCURACY

Model	Auto Prompt	User Prompt	Dongying			RS-RGB-T			VALID			BJRoad		
			mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU
AutoSAM	×	✓	96.51	96.85	94.46	88.12	91.88	81.43	85.49	90.10	77.04	80.07	87.70	70.94
	✓	×	96.07	97.67	93.98	87.29	91.06	79.38	84.79	90.62	75.64	79.08	85.02	69.48

boundary delineation. Specifically, MMoEAPM significantly accentuates target regions, and CMB facilitates richer multimodal interactions, resulting in more precise and robust segmentation outputs. These ablation results systematically demonstrate the complementary strengths of MMoEAPM, MSAFM, and CMB, and confirm that their integration is essential for achieving state-of-the-art multimodal segmentation performance with AutoSAM. Moreover, these results suggest that the Mamba-based architecture achieves higher representational efficiency and superior sequence modeling capability for multimodal data, enabling more effective integration of heterogeneous information compared to standard transformer blocks.

V. DISCUSSION

A. Analysis of Modal Data Integration

To systematically evaluate the contribution of each modality, we conducted ablation experiments by selectively removing auxiliary modal branches in AutoSAM. As reported in Table X, when AutoSAM operates with only the optical modality, there is a clear decline in segmentation accuracy across all datasets. This result highlights that AutoSAM is capable of effectively leveraging multimodal information, and the inclusion of auxiliary modalities, such as SAR, TIR, or trajectory data, consistently enhances ground perception and segmentation performance. Notably, for the RS-RGB-T dataset, the TIR modality provides the most substantial performance gain. This is primarily because TIR can compensate for the lack of discriminative information in the optical modality under adverse or nighttime conditions, a scenario frequently encountered in RS applications. These findings not only confirm the model's strong multimodal fusion capability but also provide guidance for future sensor deployment and model design in RS tasks. Beyond segmentation accuracy, we further analyze the scalability of AutoSAM in terms of resource consumption as the number of input modalities increases. The results demonstrate that both memory consumption and runtime exhibit an approximately linear growth with the number of modalities, while the total parameter count increases moderately due to the efficient design of the fusion modules. For example, enabling all modal branches on the VALID dataset increases peak GPU memory usage from 15.6 GB (single modality) to 21.7 GB (multimodal) and runtime per iteration from 0.41 to 0.48 s, while segmentation performance improves significantly. This confirms that AutoSAM maintains favorable computational efficiency and scalability, making it suitable for foundation model applications in large-scale multimodal scenarios.

B. Analysis of Prompting Paradigms

Table XI compares the effectiveness of self-prompting in AutoSAM with user-provided prompts. For this analysis, we retained all other modules and only varied the prompting strategy, evaluating both auto-prompts and user prompts derived from ground-truth labels (e.g., point and box prompts). Experimental results show that user prompts consistently yield higher segmentation accuracy, with improvements of over 1% in mIoU on the RS-RGB-T, VALID, and BJRoad datasets. This performance gap can be attributed to the inherently result-oriented nature of user prompts, which offer more accurate and targeted guidance for object localization than self-generated prompts. Nevertheless, the self-prompting still achieves competitive results, validating its practical value in fully automated RS scenarios. These observations suggest that while automated prompting is essential for scalable deployment, incorporating more precise or interactive prompt strategies can further enhance segmentation performance. Furthermore, we note that although AutoSAM currently supports point, box, and mask prompts, it does not yet utilize text prompts. The integration of text-based prompting remains an open challenge due to the lack of large-scale RS datasets with textual annotations and the inherent complexity of aligning vision-language modalities. In future work, we plan to explore the incorporation of text

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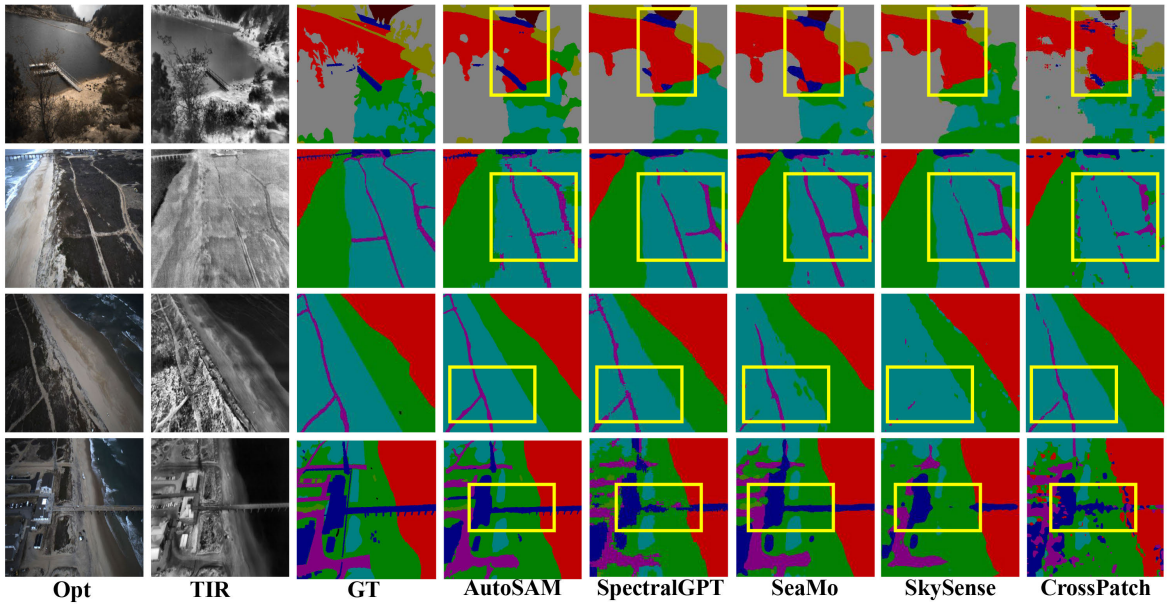


Fig. 11. Visual comparison of segmentation results on representative OPT-TIR dataset samples. Yellow boxes highlight regions where AutoSAM demonstrates superior boundary delineation and local detail preservation, especially under challenging modality conditions.

TABLE XII
COMPARISON OF VISION FOUNDATION MODELS UNDER MODALITY ROBUSTNESS SCENARIOS ON THE OPT-TIR DATASET

Type	Method	mF1 (%)	OA (%)	mIoU (%)
Unimodal	CrossPatch [77]	68.29	71.69	64.42
	SpectralGPT [78]	70.91	74.52	65.79
Multimodal	SkySense [79]	70.40	74.10	66.01
	SeaMo [45]	71.09	76.90	66.78
	AutoSAM	74.45	80.12	69.74

prompts by leveraging large vision-language models and constructing or annotating datasets with richer textual information.

C. Analysis of Modality Robustness

To further evaluate the modality robustness and generalization capacity of AutoSAM, we conducted comprehensive experiments on the OPT-TIR dataset [75], which contains challenging scenarios characterized by limited and weak TIR modality information.

As summarized in Table XII, we benchmarked AutoSAM against two state-of-the-art unimodal RS foundation models (CrossPatch [76] and SpectralGPT [77]) and two advanced multimodal models (SkySense [78] and SeaMo [45]). AutoSAM achieves the best performance on all metrics (mF1: 74.45%, OA: 80.12%, and mIoU: 69.74%), outperforming both unimodal and multimodal baselines. Notably, even under scenarios where the TIR modality provides very limited discriminative information, AutoSAM maintains strong segmentation accuracy, indicating its superior capability for cross-modal adaptation and information integration. Qualitative comparisons in Fig. 11 further illustrate the advantages of AutoSAM. The boxes highlight regions where our model produces more accurate and detailed results, especially in areas with significant modality discrepancies or complex scene structures. In contrast, unimodal models generally fail to capture critical details under such challenging conditions, likely due to the absence of complementary thermal cues. These results confirm that AutoSAM not only achieves state-of-the-art performance on conventional benchmarks, but also demonstrates remarkable robustness and scene adaptability in complex and modality-deficient scenarios.

D. Analysis of Cross-Dataset Effectiveness

To rigorously assess the generalization ability of AutoSAM, we performed comprehensive cross-dataset validation on the Dongying, RS-RGB-T, and VALID datasets, each differing significantly in class diversity and visual distribution. In this protocol, each model was trained on a single source dataset and evaluated without fine-tuning on the remaining two target datasets, reflecting a zero-shot transfer scenario. The segmentation head corresponding to each target dataset is employed during evaluation to mitigate the effects of label inconsistency. The BJRoad dataset was excluded from this analysis due to its single-class nature and substantial semantic gap from the other datasets.

As summarized in Table XIII, AutoSAM achieves the highest mIoU in 7 out of 8 cross-dataset transfer settings, with average improvements of 2.4–4.3 mIoU over the best competing baselines. For example, when trained on VALID and tested on Dongying and RS-RGB-T, AutoSAM attains mIoU scores of 78.97 and 74.41, respectively, outperforming the next-best method (RingMo-SAM) by 2.2 and 2.4 points. On average, AutoSAM yields a relative improvement of 4.8% in mIoU over the strongest baseline across all transfer scenarios. Furthermore, we observe that training on the VALID dataset, which provides the richest category diversity, consistently results in the best cross-dataset performance on the other benchmarks. In contrast, when trained on Dongying

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TABLE XIII
CROSS-DATASET VALIDATION (MIOU) ON DONGYING (A), RS-RGB-T (B), AND VALID (C) DATASETS. EACH MODEL IS TRAINED ON ONE DATASET AND EVALUATED ON THE OTHER TWO (COLUMNS B, C, AND A)

Model	Dongying (A)		RS-RGB-T (B)		VALID (C)	
	B	C	A	C	A	B
UNetFormer	53.18	36.09	64.02	35.12	68.96	55.06
Segformer	52.63	35.56	63.98	34.30	67.94	54.82
TransUNet	51.58	34.73	64.28	33.72	66.03	53.45
Deeplabv3+	51.63	34.56	63.98	34.30	65.94	54.82
RingMo-SAM	55.77	38.13	67.69	41.08	76.03	58.63
DDHRNet	52.26	35.29	65.37	37.70	70.34	57.75
V_FuseNet	53.52	36.47	65.93	37.88	71.88	56.55
FuseNet	54.41	37.44	69.08	36.50	74.88	58.06
MCANet	55.02	37.77	67.24	38.33	72.86	54.10
CMFNet	55.14	35.20	68.51	37.82	73.28	56.14
MambaDiffusion	54.63	36.27	69.89	37.06	73.88	56.86
MultiSenseSeg	56.42	37.21	69.13	40.84	72.81	57.44
FTransUNet	58.01	37.93	72.12	40.12	74.71	59.90
MMFormer	53.25	35.28	68.84	37.19	72.09	55.10
I^2HN	55.29	38.36	71.02	39.36	73.97	57.13
CFFormer	57.15	37.40	70.87	38.82	75.36	59.44
AutoSAM	59.17	40.31	74.41	45.83	78.97	63.95

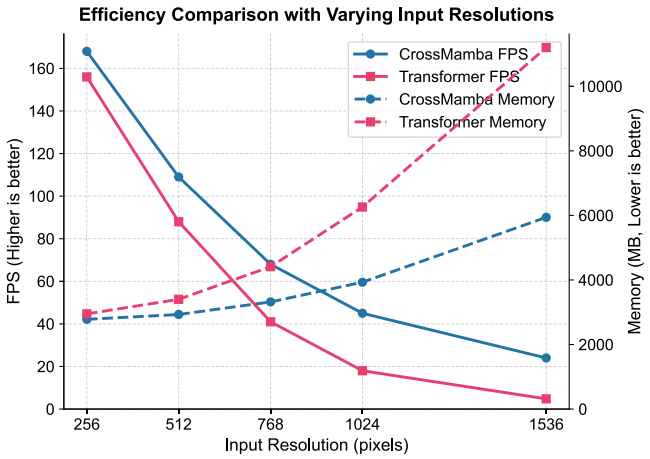


Fig. 12. Efficiency comparison across different input resolutions, showing inference speed (FPS) and GPU memory usage (MB) for CrossMamba-based and transformer-based models.

or RS-RGB-T, datasets with limited semantic coverage, all models experience substantial drops in mIoU when evaluated on VALID (e.g., AutoSAM drops to 63.95 and 58.79, respectively), highlighting the challenge of cross-domain transfer from less diverse sources. These results demonstrate the robust zero-shot and cross-dataset transfer of AutoSAM, attributable to the generalization strength of the SAM backbone and its adaptive prompting mechanism.

E. Analysis of Linear Complexity

To rigorously investigate the practical benefits of linear complexity in our framework, we performed a detailed comparison between the proposed CrossMamba and standard transformer [49] sequence modeling within AutoSAM on the RS-RGB-T dataset. As illustrated in Fig. 12, we evaluated both inference speed (FPS) and GPU memory usage across a range of input resolutions from 256 × 256 to 1536 × 1536 pixels. The CrossMamba-based model consistently maintained higher inference speeds and exhibited a much flatter degradation curve as resolution increased, demonstrating strong scalability suitable for high-resolution RS imagery.

Conversely, the transformer-based model suffered from a rapid, near-quadratic drop in FPS, highlighting the computational limitations of quadratic complexity for large-scale inputs. In parallel, GPU memory consumption for CrossMamba grew approximately linearly with input size, while the transformer variant's requirements increased steeply, exceeding 11 GB at the highest tested resolution, almost double that of CrossMamba at the same scale. These results confirm that, in real-world multimodal RS scenarios, the quadratic complexity of transformers is indeed a practical bottleneck for both speed and memory, whereas the linear-complexity CMB delivers substantial efficiency gains without sacrificing modeling capability. This empirical evidence firmly justifies our design choice and underscores the momentum for adopting efficient state-space paradigms in future foundational models for large-scale RS understanding.

VI. CONCLUSION

In this article, we presented AutoSAM, an end-to-end visual foundation model tailored for multimodal RS semantic segmentation. AutoSAM incorporates a MoE framework and dictionary learning via the proposed MMoEAPM module, which effectively addresses the reliance of SAM on external prompts in multimodal scenarios. Notably, MMoEAPM is designed as a plug-and-play strategy, thus preserving user interaction flexibility. We further introduced the CMB, leveraging pairwise alternating sequence scanning to enable efficient contextual fusion and global information capture across modalities. In addition, the integration of the MSAFM module compensates for the limited detail-capturing capability of conventional SAM architectures, significantly improving segmentation accuracy for small-scale targets. Extensive experiments on diverse datasets, encompassing both real-world and synthetic multimodal scenes, demonstrate the superior performance and versatility of AutoSAM. To our knowledge, AutoSAM represents the first self-prompting foundation model in multimodal RS, capable of robustly handling a wide range of modality combinations without external prompts. Despite these advances, challenges remain regarding model efficiency and deployment in practical applications. Future research will explore adaptive cross-modal alignment, spatio-temporal change perception, and knowledge-enhanced foundation modeling, further improving robustness and generalization in complex RS scenarios.

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